

**GEONARRATIVE AI: AN INTEGRATED GIS AND 3D DIGITAL TWIN
FRAMEWORK FOR URBAN FLOOD SUSCEPTIBILITY ASSESSMENT
AND DECISION SUPPORT FOR PUNE CITY**

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Certificate

This is to certify that the project report entitled "GeoNarrative AI: An Integrated GIS and 3D Digital Twin Framework for Urban Flood Susceptibility Assessment and Decision Support For Pune City" submitted by Savant Harnish Sanjaybhai (PRN: 25070243032) is a bonafide record of the research work carried out under my supervision in partial fulfilment of the requirements for the award of the degree of Master of Science in Data Science and Spatial Analytics at Symbiosis Institute of Geoinformatics, Symbiosis International (Deemed University), Pune.

The results embodied in this dissertation have not been submitted to any other university or institution for the award of any degree or diploma, to the best of my knowledge and belief.

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Declaration

I hereby declare that the dissertation entitled "GeoNarrative AI: An Integrated GIS and 3D Digital Twin Framework for Urban Flood Susceptibility Assessment and Decision Support in Pune" is an authentic record of my own work carried out as a part of the MSc programme in Data Science and Spatial Analytics at Symbiosis Institute of Geoinformatics, Symbiosis International (Deemed University), Pune, under the guidance of Dr. T. P. Singh.

I further declare that this work has not been submitted, either in part or in full, for any other degree, diploma, or award at this or any other university or institution. All sources of information and assistance used during this research have been duly acknowledged.

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Abstract

Urban flooding poses a recurrent and severe disaster threat to rapidly developing metropolitan areas in monsoon-prone regions. In Pune, situated at the confluence of the Mula and Mutha rivers in western Maharashtra, heavy precipitation events and rapid surface sealing have repeatedly triggered severe inundation, affecting vulnerable populations and transport lifelines. While standard Geographic Information System (GIS) modelling identifies susceptible topography, conventional static risk maps lack three-dimensional spatial context, fail to communicate dynamic temporal flood progression, and require specialized GIS expertise to interpret, thereby limiting their utility for operational disaster management.

To bridge this critical socio-technical and interdisciplinary decision-support gap, this study develops GeoNarrative AI, an integrated spatial decision-support framework for the Pune Municipal Corporation (PMC) administrative extent (506.91 km²). The methodology systematically unifies three pillars: (1) a multi-criteria GIS flood susceptibility assessment utilizing an AHP-informed Weighted Linear Combination (WLC) over conditioning factors derived from 30m Copernicus DEM and Sentinel-2 LULC datasets; (2) a terrain-constrained temporal flood scenario engine coupled with an interactive 3D Urban Digital Twin developed via CesiumJS, incorporating 180,307 municipal building footprints and 55,309 road network segments (7,445.90 km total length); and (3) an artificial intelligence layer featuring domain-grounded Natural Language Geospatial Interaction (GeoAI Assistant) and an automated Geospatial Technical Report Agent.

Quantitative exposure results across simulated scenario-based flood severity levels demonstrate a monotonic expansion in inundation impact: from 53.60 km² of temporary flood extent affecting 11,262 building footprints (8,808 within critical riparian zones) and 751.19 km of road segments under Normal flood conditions, scaling progressively up to 133.97 km² of inundation impacting 40,723 building footprints (32,084 critical) and 1,877.47 km of road segments under Extreme flood scenarios. The integration of live spatial analytical dashboards with project-aware conversational GeoAI enables decision-makers to interrogate geospatial models via natural language without software barriers.

The outcomes affirm that uniting multi-criteria spatial analysis, temporal 3D digital twin visualization, and generative spatial intelligence creates an accessible, highly interpretive disaster decision-support instrument. This research advances disaster communication by transforming static spatial analytical outputs into dynamic, intelligible, and actionable risk intelligence.

Keywords: Urban Flood Susceptibility, 3D Digital Twin, AHP-Informed Weighted Overlay, GeoAI, Natural Language Geospatial Interaction, CesiumJS, PostGIS, Spatial Decision Support System, Pune Municipal Corporation.

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List of Abbreviations

Abbreviation	Full Form
AHP	Analytical Hierarchy Process
WLC	Weighted Linear Combination
API	Application Programming Interface
CRS	Coordinate Reference System
DEM	Digital Elevation Model
DSS	Decision Support System
DT	Digital Twin
EPSG	European Petroleum Survey Group
ESA	European Space Agency
FSI	Flood Susceptibility Index
FSM	Flood Susceptibility Mapping
GeoAI	Geospatial Artificial Intelligence
GIS	Geographic Information System
IMD	India Meteorological Department
KPI	Key Performance Indicator
LLM	Large Language Model
LULC	Land Use / Land Cover
MCDA	Multi-Criteria Decision Analysis
MCE	Multi-Criteria Evaluation
NLP	Natural Language Processing
OSM	OpenStreetMap
PMC	Pune Municipal Corporation
PMRDA	Pune Metropolitan Region Development Authority
REST	Representational State Transfer

SDSS	Spatial Decision Support System
SRTM	Shuttle Radar Topography Mission
UI / UX	User Interface / User Experience
UTM	Universal Transverse Mercator
WGS	World Geodetic System

Chapter 1: Introduction

1.1 Background

Urban flooding ranks among the most economically damaging and socially disruptive natural hazards in monsoon-prone regions of South Asia. Rapid surface sealing driven by urbanization, combined with increasingly intense precipitation events associated with climatic variability, systematically diminishes urban infiltration capacity and elevates surface runoff rates. In major metropolitan centers such as Pune in western Maharashtra, heavy monsoon downpours intersecting with dense urban form have repeatedly triggered severe inundation events, resulting in infrastructure disruption, financial loss, and socio-economic vulnerability.

Geographic Information System (GIS) flood susceptibility mapping (FSM) serves as an established analytical technique for identifying spatial zones inherently prone to inundation based on topographic, hydrological, and anthropogenic conditioning factors. Among multi-criteria decision analysis (MCDA) techniques, the Analytical Hierarchy Process (AHP) weighted overlay method has been widely utilized because it enables rigorous spatial stratification of hazard zones without relying exclusively on comprehensive historical hydrographs or dense sensor networks. By incorporating factors such as elevation, terrain slope, proximity to drainage channels, and land cover impermeability, AHP-based FSM provides valuable diagnostic spatial intelligence.

Despite analytical progress in spatial hazard modelling, conventional flood susceptibility assessments consistently produce static, two-dimensional outputs such as static maps or offline reports. These representations exhibit significant limitations in operational decision-making. Static maps lack temporal dynamism, fail to communicate how floodwater propagates across complex three-dimensional urban topography, and cannot interactively quantify critical infrastructure exposure under varying severe rainfall scenarios. Furthermore, technical proficiency barriers associated with specialized desktop GIS packages impede municipal authorities, disaster managers, and planners from directly interrogating analytical hazard layers during critical planning or disaster mitigation workflows.

The emergence of interactive 3D Urban Digital Twins and Generative Geospatial Artificial Intelligence (GeoAI) presents an opportunity to overcome these classical GIS limitations. A Digital Twin integrates structural geospatial data with simulation engines to provide an explorative virtual replica of urban infrastructure. By synthesizing multi-criteria flood susceptibility modeling, terrain-constrained temporal inundation scenarios, building-level exposure analytics, and domain-grounded conversational AI into a unified web platform, this research establishes an advanced Spatial Decision Support System (SDSS) tailored for the Pune Municipal Corporation.

1.2 Problem Statement

The overarching problem addressed by this research is the critical socio-technical disconnection between complex GIS-based flood hazard models and practical urban decision-making. Specifically, the research argument evolves through the following sequence of systemic limitations:

- **Urban Flood Vulnerability:** Monsoon-driven precipitation intersecting with dense impervious surfaces creates high flood vulnerability in urban centers like Pune.
- **Complex Spatial Interaction:** Inundation risk emerges from complex, non-linear interactions between topographic elevation, terrain slope, river hydrology, and urban built-up density.
- **Identification via FSM:** While GIS-based multi-criteria susceptibility modeling successfully identifies vulnerable terrain, conventional methodologies culminate solely in static diagnostic outputs.
- **Static Interpretational Barriers:** Two-dimensional static maps capture only a single static hazard layer, offering limited temporal interpretability and completely lacking three-dimensional visual intuition of urban morphology.
- **Obfuscated Infrastructure Exposure:** In traditional workflows, quantifying the precise vulnerability of urban lifelines—such as specific residential building footprints or transportation corridors—under progressive scenario-based flood severity levels requires tedious, manual spatial intersection procedures.

- **Fragmented Analytical Ecosystem:** Quantitative statistics, GIS hazard layers, temporal scenario simulations, and explanatory reports typically reside in decoupled software tools, hindering rapid synthesis during urban planning or disaster preparedness.
- **Need for Integrated Decision Support:** There exists an acute analytical requirement for an accessible, interoperable environment that reconciles spatial hazard modeling with dynamic exploratory visualization.
- **The GeoNarrative Framework:** Consequently, this research develops the GeoNarrative AI framework, bridging advanced spatial analytical modeling with 3D digital twin simulations, natural-language geospatial query capabilities, and automated decision intelligence.

1.3 Research Gap

An evaluation of contemporary literature reveals that GIS flood susceptibility modeling, 3D urban digital twins, and artificial intelligence decision systems have historically evolved within siloed disciplinary paradigms. Numerous empirical studies implement AHP weighted overlays or statistical GIS modeling for flood mapping, yet their intellectual contribution typically halts at static cartographic validation within desktop software. Conversely, modern Digital Twin frameworks frequently excel at visual geometry rendering or energy monitoring, but rarely incorporate formal geospatial disaster multi-criteria models or dynamic vector exposure algorithms directly within the visualization canvas.

Crucially, a definitive gap exists in computational integration and analytical interpretability. While recent studies explore web-based GIS dashboards (Level 2 Diagnostic Digital Twins), there remains a pronounced deficit in frameworks that unite Level 3 Simulation Digital Twins with Natural Language Geospatial Interaction (GeoAI). Specifically, the reviewed literature revealed limited evidence of frameworks within Indian municipal disaster-planning contexts that simultaneously integrate multi-criteria flood susceptibility grids, temporal terrain-constrained flood scenario propagation, real-time vector infrastructure exposure intersections, and domain-grounded conversational AI to improve the accessibility of spatial insights. This dissertation directly targets this defensible integration, interpretability, and decision-support gap.

1.4 Research Questions

To guide the conceptual and computational development of this research, three core research questions were established:

- RQ1: How can multi-criteria GIS spatial conditioning factors be integrated to delineate relative flood susceptibility zones across Pune Municipal Corporation?
- RQ2: How can scenario-driven temporal flood propagation be modeled and coupled with spatial vector intersections to quantify progressive infrastructure exposure under varying scenario severity levels?
- RQ3: How can an integrated 3D Urban Digital Twin and AI-assisted geospatial decision-support environment support the interpretation, communication, and exploration of spatial flood intelligence?

1.5 Research Aim and Objectives

The primary research aim of this dissertation is to architect, implement, and evaluate an integrated GIS, 3D Digital Twin, and AI-assisted geospatial decision-support framework (GeoNarrative AI) for urban flood susceptibility assessment and infrastructure exposure modeling in Pune.

To achieve this aim, three specific, measurable research objectives are defined:

- Objective 1: To construct and classify a multi-criteria GIS flood susceptibility model for the Pune Municipal Corporation area utilizing high-resolution topographic and remote sensing conditioning factors.
- Objective 2: To simulate terrain-constrained temporal flood propagation scenarios and computationally evaluate quantitative exposure across municipal building footprints and transportation networks.
- Objective 3: To integrate spatial analytics within a browser-based 3D Digital Twin featuring domain-grounded conversational GeoAI and automated reporting to support accessible, exploratory disaster risk decision-making.

A systematic methodological alignment connecting research questions, objectives, methodological techniques, results, and thesis conclusions is presented in Table 1.1.

Table 1.1: Research Question, Objective, and Methodology Alignment Matrix

Research Question	Specific Objective	Methodological Technique	Analytical Result	Conclusion Trace
RQ1: Spatial susceptibility conditioning formulation	Obj 1: Construct & classify multi-criteria FSM	AHP-informed weighted overlay (Copernicus DEM, Sentinel-2 LULC, distance, slope)	5-class FSI mapping & municipal susceptibility zones (Ch 6.7)	Spatial hazard stratification with qualitative coherence assessment (Ch 7.3)
RQ2: Scenario temporal propagation & infrastructure exposure	Obj 2: Simulate temporal flood expansion & infrastructure impact	Terrain-constrained flood expansion algorithm & vector spatial intersection	Monotonic progression statistics, building/road exposure tables (Ch 6.8)	Quantified scenario vulnerability benchmarks (Ch 7.3)
RQ3: Integrated 3D Digital Twin & AI decision support	Obj 3: Develop web 3D twin with GeoAI interaction & reporting	CesiumJS tiling, PostGIS/FastAPI pipeline, LLM context tool routing	Interactive command dashboard, GeoAI assistant, technical report agent (Ch 6.9)	Accessible spatial decision intelligence (Ch 7.3)

1.6 Scope and Limitations

The operational boundaries and methodological parameters governing this research are defined as follows:

- **Geographic Extent:** The study strictly encapsulates the administrative territory of the Pune Municipal Corporation (PMC) in Maharashtra, India, spanning an analytical area of approximately 506.91 km² bounded within longitudes 73.7319°E to 74.0184°E and latitudes 18.3854°N to 18.6218°N.
- **Spatial Datasets:** Primary raster datasets include the Copernicus GLO-30 Digital Elevation Model (~30m horizontal resolution) and European Space Agency (ESA) Sentinel-2 derived Land Use / Land Cover (LULC, 10m resolution). Vector geodatabase

layers incorporate 180,307 municipal building footprints and 55,309 road network segments derived from verified GIS survey databases.

- **Analytical Hazard Scope:** Susceptibility analysis relies upon an AHP-informed Weighted Linear Combination (WLC) multi-criteria evaluation, using criterion weights adapted from published literature rather than an independently elicited pairwise-comparison matrix. Due to computational efficiency requirements and the absence of micro-scale underground stormwater network schematics, 2D hydrodynamic finite-element simulation (e.g., fully calibrated HEC-RAS 2D) is explicitly excluded.
- **Scenario & Exposure Limitations:** Inundation progression utilizes terrain-constrained spatial expansion rather than fluid dynamic Navier-Stokes modeling. Reported water depth statistics reflect scenario-derived relative inundation depth estimates over DEM cells rather than field-calibrated physical gauge depths.
- **Validation Framework:** Due to historical flood extent data scarcity, methodological validation relies upon spatial coherence checks against documented 2019/2024 riparian inundation corridors, scenario monotonicity verification, and programmatic infrastructure intersection validation.

1.7 Contribution of the Study

This thesis makes a rigorous academic contribution at the intersection of Spatial Data Science, Geoinformatics, and Disaster Decision Support. Specifically, the contributions comprise:

- **Methodological Integration:** A reproducible geospatial pipeline that moves from static GIS AHP-informed weighted susceptibility analysis to interactive, temporal 3D scenario simulations.
- **Quantitative Municipal Infrastructure Assessment:** Verified numerical exposure evaluations quantifying building footprint containment (11,262 to 40,723 affected units) and road segment inundation (751.19 km to 1,877.47 km) across four scenario-based flood severity levels.
- **Natural Language Geospatial Intelligence:** The implementation of a project-aware conversational GeoAI Assistant and automated Report Agent capable of real-time

geospatial tool execution, lowering software proficiency barriers for non-specialist decision-makers.

1.8 Organisation of the Dissertation

This dissertation is systematically structured across seven chapters. Chapter 2 synthesizes theoretical literature regarding urban flood susceptibility, multi-criteria spatial modeling, Digital Twins, and GeoAI systems. Chapter 3 introduces the geographic, climatic, and hydrological profile of the Pune Municipal Corporation study area along with data catalog characteristics. Chapter 4 delineates the scientific research methodology, detailing pre-processing, AHP-informed weighted-overlay mathematical formulation, scenario propagation algorithms, and spatial vector intersections. Chapter 5 covers system software engineering, data pipeline architecture, and 3D web rendering integration. Chapter 6 presents verified quantitative analytical results across GIS hazard layers, scenario exposure models, and interactive module performances along with scholarly discussion. Finally, Chapter 7 summarizes key research conclusions, traces objective achievements, addresses methodological limitations, and outlines recommendations for future research.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the published literature across three domains that constitute the theoretical foundation of this study: flood susceptibility modelling using GIS-based multi-criteria methods, Digital Twin applications in urban systems and disaster management, and web-based GIS platforms for spatial decision support. The review identifies specific gaps that motivated the design choices adopted in this project.

2.2 Flood Susceptibility Modelling

Flood susceptibility mapping identifies areas inherently prone to inundation based on geographic, hydrological, and land-use characteristics. The literature broadly distinguishes between data-driven machine learning methods and knowledge-driven multi-criteria approaches (Tehrany et al., 2019).

2.2.1 Machine Learning Approaches

Rahmati et al. (2016) applied Support Vector Machines and Random Forest classifiers to predict flood-prone areas in the Golestan Province, Iran, using 12 conditioning factors. Their study found that Random Forest outperformed SVM with an area under the curve (AUC) of 0.89. Zhao et al. (2019) compared Logistic Regression, SVM, and deep neural networks for flash flood susceptibility in a mountainous catchment, reporting that ensemble methods produced the most stable predictions. While these machine learning approaches achieve high predictive accuracy, they require labelled training data in the form of historical flood extent polygons. Such records are often unavailable or unreliable for Indian metropolitan areas, limiting the direct applicability of data-driven methods in data-scarce settings.

2.2.2 Multi-Criteria Decision Analysis

Multi-criteria decision analysis (MCDA) methods circumvent the labelled data requirement by relying on expert-assigned weights for spatial conditioning factors. The Analytical Hierarchy Process (AHP) developed by Saaty (1980) is the most widely adopted MCDA method in flood susceptibility research. Ouma and Tateishi (2014) applied AHP-weighted

overlay analysis in a GIS environment to map flood vulnerability in Kenya, combining slope, elevation, drainage density, and soil permeability. Their approach achieved reasonable concordance with observed flood events and has since been replicated across monsoon-prone regions.

Samanta et al. (2018) applied a similar AHP framework for flood susceptibility mapping in the Ajay River basin, West Bengal, using drainage density, rainfall, slope, soil type, and land use as conditioning factors. Their study is notable for documenting the consistency ratio of the AHP pairwise comparison matrix, a validation step frequently omitted in other studies. Souissi et al. (2020) combined AHP with fuzzy logic in Tunisia, demonstrating that hybrid MCDA techniques can accommodate uncertainty in weight assignment.

A common limitation across these studies is that the weighted overlay computation is performed within desktop GIS software, producing a static raster output. The model runs once, the result is exported as an image, and any modification to weights or input data requires repeating the entire analysis manually. This static workflow limits the practical utility of the susceptibility map for ongoing urban planning decisions.

2.2.3 Conditioning Factors in Urban Environments

The selection of conditioning factors for flood susceptibility varies across studies but consistently includes elevation, slope, and proximity to watercourses as primary factors (Tehrany et al., 2019). In urban environments, additional factors become relevant: building density, which correlates with surface imperviousness, and land use and land cover, which determines infiltration capacity. This study incorporates five conditioning factors selected to capture both natural topographic influences and urban-specific impermeability characteristics.

2.3 Digital Twin Technology

The Digital Twin concept was formalised by Grieves (2014) as comprising a physical space, a virtual space, and the data connections between them. In urban applications, Digital Twins integrate heterogeneous spatial data within a three-dimensional virtual environment that mirrors the physical city.

2.3.1 Urban Digital Twins

Bolton et al. (2018) established the Gemini Principles for national Digital Twins, emphasising openness, security, and federation. Dembski et al. (2020) proposed an urban Digital Twin framework that integrates IoT sensors, BIM models, and GIS layers, distinguishing between static assets (terrain, buildings), semi-dynamic assets (infrastructure), and dynamic streams (traffic, weather). Their classification directly informs the asset categorisation adopted in this study, where terrain and buildings are treated as persistent layers while flood water and weather effects are rendered dynamically.

2.3.2 Digital Twins in Flood Management

White et al. (2021) reviewed Digital Twin applications in smart cities and identified three maturity levels. Level 1 systems provide static descriptive GIS. Level 2 systems offer diagnostic analytical dashboards. Level 3 systems integrate predictive capabilities and simulation. Most existing flood management platforms operate at Level 1 or Level 2, presenting pre-computed maps without analytical or simulation capabilities. This study targets a Level 2 to Level 3 system by coupling GIS-based flood susceptibility analysis with animated scenario simulation within a 3D environment.

Ford and Wolf (2020) examined how Digital Twins can support disaster management through real-time data integration and scenario planning. Their work highlights that the value of a Digital Twin lies not only in visual realism but in enabling decision-makers to interact with spatial information rather than passively consuming static outputs.

2.4 Web-Based GIS for Decision Support

The transition from desktop to web-based GIS has been accelerated by client-side mapping libraries and cloud-hosted terrain services. Yin et al. (2019) developed a web-based flood risk communication platform that demonstrated interactive visualisation improves stakeholder understanding compared to static PDF maps. However, their system served pre-computed layers without embedding analytical capabilities. Iqbal et al. (2021) built a web GIS for urban flood monitoring in Dhaka incorporating spatial queries and interactive map rendering, but without 3D terrain visualisation or scenario simulation.

CesiumJS, the library adopted in this study, supports 3D terrain rendering with accurate elevation data, global building geometry from OpenStreetMap, and GPU-accelerated water material rendering. These capabilities make CesiumJS well-suited for Digital Twin applications that require three-dimensional spatial context for flood visualisation (Cesium, 2025).

2.5 Research Gap Analysis

Table 2.1 presents a comparative analysis of the capabilities offered by existing approaches against those implemented in this study.

Table 2.1: Comparative Analysis of Flood Decision-Support & Visualization Frameworks

Capability	Desktop GIS	Web Flood Platforms	This Study
Flood susceptibility mapping	Yes (static)	Pre-computed only	Yes (GIS-based)
3D terrain visualisation	Limited	Some (2D dominant)	Yes (CesiumJS)
Interactive scenario simulation	No	Limited	Yes (4 scenarios)
Building-level 3D rendering	No	No	Yes (OSM Buildings)
Animated flood progression	No	No	Yes (terrain-aware)
Web accessible	No (desktop)	Yes	Yes
Decision support dashboard	No	Partial	Yes (live telemetry)
Conversational GeoAI copilot	No	No	Yes (Gemini LLM tools)
Automated technical report agent	No	No	Yes (real-time studio)
Open-source spatial stack	No (proprietary)	Varies	Yes

The comparison reveals that while desktop GIS provides robust analytical capabilities for flood susceptibility mapping, it lacks interactive visualization, temporal dynamism, and web accessibility. Existing web flood platforms provide some interactivity but do not embed analytical multi-criteria models or generative artificial intelligence within a 3D environment. This study bridges this socio-technical gap by integrating a GIS-based AHP-informed flood susceptibility model with an interactive Level 3 Digital Twin, conversational GeoAI co-pilot capabilities, and real-time automated reporting within a single web-accessible platform.

2.6 Chapter Summary

This chapter reviewed the relevant literature across flood susceptibility modelling, Digital Twin technology, and web-based GIS. The review established that AHP-based weighted overlay is a well-validated method for flood susceptibility assessment in data-scarce environments, that Digital Twin technology offers compelling advantages for interactive flood visualisation, and that the integration of these two approaches within a single platform remains an underexplored area. The next chapter describes the study area and its geographic, hydrological, and demographic characteristics.

Chapter 3: Study Area and Data Catalog

3.1 Introduction

The precise geographical bounding of a study area and rigorous inventorying of corresponding spatial datasets form the foundation of reproducible geospatial analytical research. The physical characteristics of the terrain—comprising topographic variation, hydrological drainage patterns, meteorological regimes, and anthropogenic impervious surfaces—directly dictate the parametric formulations of flood susceptibility models and exposure algorithms. This chapter characterizes the geographic, meteorological, hydrological, and demographic profile of the Pune Municipal Corporation (PMC) administrative extent. It also details the technical metadata of all primary raster and vector geospatial datasets curated for this research.

3.2 Location and Administrative Extent

The city of Pune is located in the western sector of the state of Maharashtra, India, situated on the leeward plateau of the Sahyadri mountain range (Western Ghats). While the broader urban agglomeration encompassing the Pune Metropolitan Region Development Authority (PMRDA) spans extensive territory, the analytical extent of this study is strictly confined to the official administrative boundary of the Pune Municipal Corporation (PMC).

The finalized PMC study area covers a computed geographical footprint of approximately 506.91 km². In spatial coordinate space, the municipal extent lies within geodetic bounding latitudes of 18.3854°N to 18.6218°N and longitudes of 73.7319°E to 74.0184°E (referenced to World Geodetic System 1984, EPSG:4326). The municipal boundary vector polygon, curated from verified administrative geodatabases (PMC.geojson), serves as the definitive spatial clipping mask and analytic framework for all multi-criteria evaluations and infrastructure intersections.

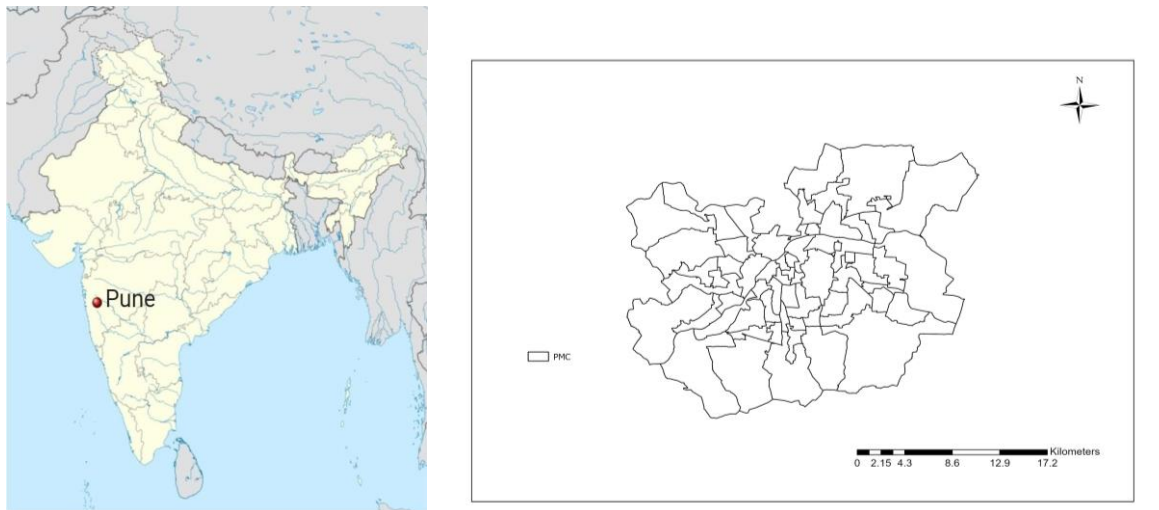


Figure 3.1: Geographic location and administrative boundary of Pune Municipal Corporation (506.91 km²) within Maharashtra, India.

3.3 Climate and Meteorological Regime

Pune falls within a tropical wet-and-dry climatic regime (Köppen climate classification Aw), exhibiting three distinct seasons: a high-temperature pre-monsoon summer from March to May, the southwest monsoon season extending from June to September, and a moderate winter from October to February. The southwest monsoon dominates the annual regional hydrological cycle, contributing approximately 85% of the annual precipitation, which averages 722 mm across the urban basin but regularly exceeds 1,100 mm during intense cyclonic or active monsoon years.

Precipitation distribution across the region displays pronounced spatial heterogeneities. Western catchments positioned adjacent to the topographical barrier of the Western Ghats receive markedly higher atmospheric moisture deposition due to strong orographic uplift. Consequently, extreme rainfall occurrences in western up-country basins generate rapid overland storm volumes that drain eastward directly through the urban channels of Pune.

Recent climate patterns indicate a quantifiable shift toward short-duration, extreme precipitation bursts. Notable historical disaster events—such as the intense cloudburst and river flooding of September 2019 (exceeding 200 mm precipitation within 24 hours) and severe pluvial inundation events in July–August 2024—resulted in severe transport

disruptions, infrastructure loss, and evacuation of riparian communities. These meteorological characteristics point to the need for scenario-driven simulation capabilities capable of representing extreme rainfall intensities.

3.4 Hydrological Network and Riparian Morphology

The natural surface hydrology of Pune is structurally defined by the Mula-Mutha river system and its dense secondary tributary channels. The Mutha river enters the urban extent from the west (regulated by the upstream Khadakwasla and Mulshi dam reservoirs), while the Mula river penetrates from the northwest (regulated by the Pavana and Mulshi alignments). The two prominent waterways converge at Sangamwadi within the urban center, forming the joint Mula-Mutha river, which subsequently traverses eastward toward the Bhima river basin.

For this study, hydrological features were extracted from verified hydrographic layers, yielding 171 primary permanent water body and stream channel polygon features within the PMC boundary. Combined, these permanent aqueous features account for an observed surface water baseline area of approximately 9.06 km². Proximity to these 171 channels constitutes a major weighting criterion in the analytical susceptibility framework, representing direct lateral exposure to riparian overflow during peak dam discharges or high-volume rainfall runoff.

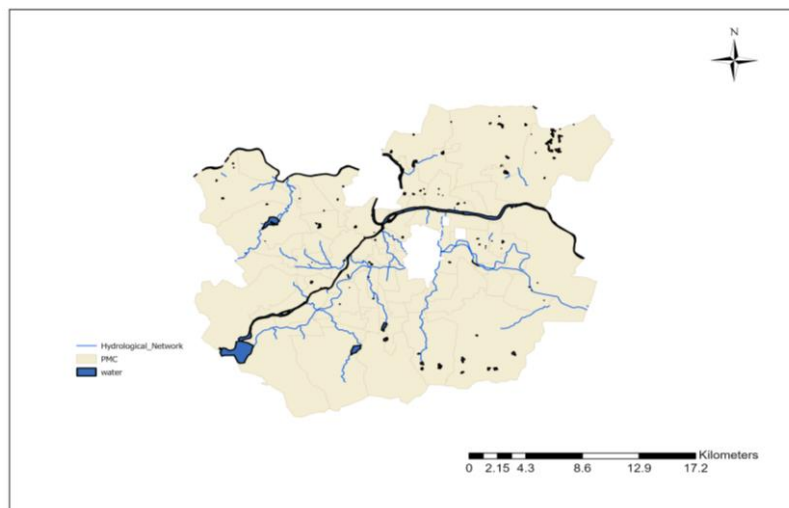


Figure 3.2: Spatial mapping of permanent Mula-Mutha hydrological corridors and primary municipal water features (171 features spanning 9.06 km²).

3.5 Topographic Profile and Terrain Morphology

The topographic landscape of the PMC area exhibits significant geomorphological variance. Elevation values ascend rapidly toward western hills and southern ridgelines, reaching altitudes upwards of 820 meters above mean sea level (MSL). Conversely, the central urban valley and Mula-Mutha riparian plains transition across relatively planar alluvial terrain at elevations between 530 and 560 meters MSL.

To model these physical variations mathematically, digital surface topography was procured from the Copernicus GLO-30 Digital Elevation Model, offering global standardized coverage at approximately 30-meter horizontal spatial resolution. Parametric raster surface derivatives—specifically continuous slope gradients (degrees) and topographic curvature—were computed directly from the Copernicus elevation matrix. Low-lying planar valley formations exhibit minimal hydraulic gravitational drainage momentum, forming primary spatial accumulation zones for pluvial ponding and overbank floods.

Table 3.1: Summary of Topographic and Geomorphological Parameters in the PMC Study Area

Terrain Parameter	Value Range	Geomorphological & Hydrological Relevance
Elevation Altitude	530 – 820+ meters MSL	Lowest elevations define natural accumulation basins and floodplain limits.
Slope Gradient	0.0° – 35.0°+	Planar terrain (<2.5°) promotes pluvial stagnation; steep slopes (>15°) accelerate flash runoff.
Riparian Corridor	530 – 555 meters MSL	Primary fluvial discharge valley bounding central municipal wards.

3.6 Land Use / Land Cover (LULC) and Surface Impermeability

Over the past three decades, the PMC territory has experienced expansive demographic and structural transformation driven by rapid industrial growth and commercial information technology development. This rapid urban expansion has systematically converted historical

agricultural land, vegetated natural scrublands, and riparian buffers into dense impervious built environments.

To evaluate urban impermeability, high-resolution remote sensing Land Use / Land Cover (LULC) raster arrays were extracted at 10-meter pixel resolution from European Space Agency (ESA) Sentinel-2 multi-spectral classifications. Built-up surfaces dramatically suppress soil hydraulic infiltration capacity, resulting in localized surface runoff coefficients approaching 0.85 to 0.95. This rapid transformation of rainfall into surface runoff substantially intensifies peak discharges entering urban drainage networks.

3.7 Demographics and Urban Vulnerability

The population residing within the Pune municipal territory has scaled rapidly, growing from approximately 3.1 million in the 2011 decennial census to an estimated population exceeding 5.2 million across the expanded PMC jurisdiction by 2025. Demographic density within older central urban wards (such as Kasba Peth, Sadashiv Peth, and Shivajinagar) exceeds 25,000 persons per square kilometer. High residential density positioned within low-elevation riparian zones increases socio-economic vulnerability during sudden inundation episodes, requiring robust spatial decision tools.

3.8 Curated Geospatial Data Catalog

A comprehensive spatial geodatabase was compiled by processing multi-source remote sensing arrays and vector structural surveys. To ensure total scientific reproducibility, the vector layers representing urban lifelines and structural buildings were curated from comprehensive municipal surveys (MyProject8.gdb) and precisely clipped to the definitive PMC administrative boundary.

The structural building inventory originally encompassed 339,732 regional source features; post-clipping to the PMC operational boundary yielded an analytical dataset of exactly 180,307 municipal building footprints covering an aggregate built structural area of approximately 42.92 km². Similarly, the transportation linear network layer comprises exactly 55,309 road network segments, spanning a cumulative operational distance of

7,445.90 kilometers within PMC boundaries. A detailed specification of the integrated dataset catalog is summarized in Table 3.2.

Table 3.2: Technical Specifications of Curated Geospatial Data Catalog for Pune Municipal Corporation

Dataset Element	Source Institution / Platform	Resolution / Scale	Verified Analytical Metrics (Inside PMC)
Administrative Boundary	PMC GIS / OpenStreetMap	Vector Polygon	506.91 km ² territorial extent (PMC.geojson)
Digital Elevation Model	Copernicus GLO-30 (ESA)	30-meter Grid Raster	Continuous elevation matrix (530m to 820m)
LULC Classification	Sentinel-2 WorldCover (ESA)	10-meter Pixel Raster	Multi-class surface impermeability categorization
Building Footprints	MyProject8.gdb / PMC Surveys	Object Vector Polygon	180,307 building structures (~42.92 km ² footprint area)
Road Network Segments	OpenStreetMap / PMC Routing	Vector Polyline	55,309 roadway segments (7,445.90 km total length)
Permanent Hydrology	Hydrographic Vector Database	Vector Polygon / Line	171 water bodies and rivers (9.06 km ² permanent water area)

3.9 Justification for Study Area Selection

The selection of the Pune Municipal Corporation extent as the analytical testbed is justified by several academic and practical considerations:

- Hydrological Hazard Severity: Recent severe monsoon flooding events (2019, 2024) provide documented empirical evidence of severe vulnerability across low-lying municipal sectors.

- **Geomorphological Complexity:** The juxtaposition of steep western hills with planar central floodplains presents an ideal topographic testing environment for evaluating multi-criteria spatial conditioning models.
- **Data Quality and Completeness:** High-resolution Copernicus terrain grids combined with comprehensive structural building footprint polygons (180,307 geometries) enable precise vector-raster computational intersections without relying upon estimated proxy formulas.
- **Policy and Smart City Relevance:** As a prominent municipal administration undergoing rapid Smart City transformations, PMC derives immediate operational value from integrating interactive 3D digital twins and natural language GeoAI decision frameworks.

3.10 Chapter Summary

This chapter established the geographical, topographical, and hydrological characteristics of the Pune Municipal Corporation study area (506.91 km²). It also cataloged the analytical datasets engineered into the project geodatabase, verifying exact building counts (180,307 units), transportation segments (55,309 lines spanning 7,445.90 km), and hydrological water features (171 bodies, 9.06 km²). These verified spatial layers serve as the foundational inputs for the analytical methodologies detailed in Chapter 4.

Chapter 4: Research Methodology

4.1 Introduction

This chapter articulates the rigorous scientific methodology structured to design, evaluate, and deploy the GeoNarrative AI geospatial decision-support framework. Transitioning beyond conventional static GIS cartography, this research adopts an integrated analytical workflow connecting multi-criteria decision analysis (MCDA), spatial vector intersection algorithms, terrain-constrained scenario propagation, three-dimensional digital twin rendering, and natural language generative intelligence. Every procedural step—from coordinate harmonization to AI tool routing—is formally specified to ensure complete scientific reproducibility.

4.2 Methodological Workflow Framework

The overarching methodology executes across seven interdependent analytical phases as illustrated in the publication-quality methodological framework diagram (Figure 4.1). The pipeline systematically progresses from problem conceptualization and data acquisition through spatial conditioning and susceptibility indexing, culminating in temporal simulation, interactive 3D digital twin orchestration, and artificial intelligence decision support.

[DIAGRAM: Research Methodology Framework]

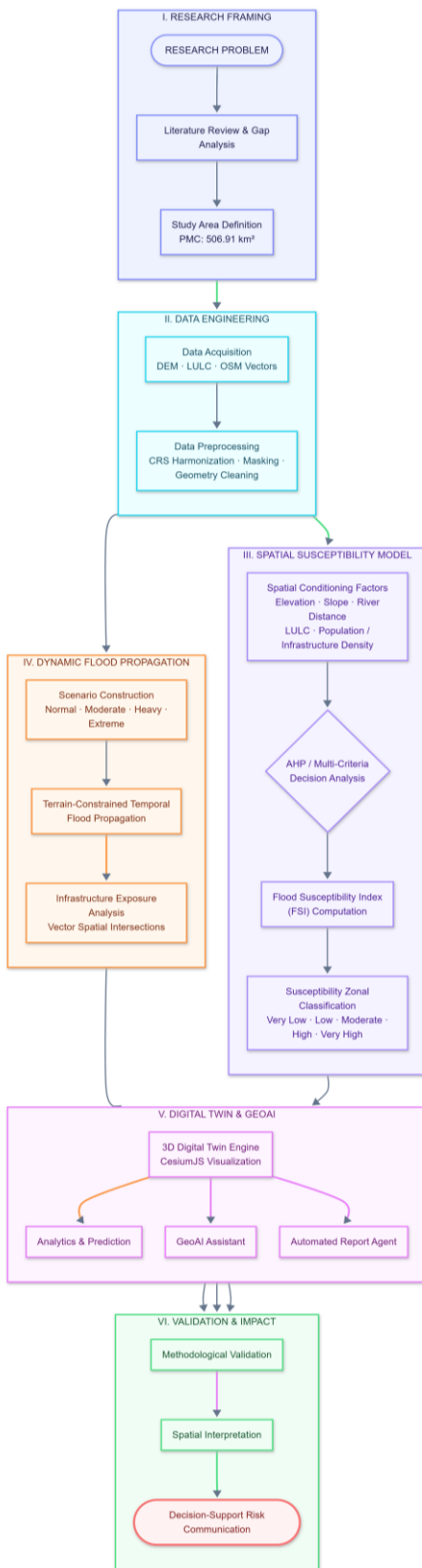


Figure 4.1: Methodological Workflow Framework of GeoNarrative AI

4.3 Data Preprocessing and Spatial Harmonization

Raw multi-source spatial datasets inherently exhibit divergent coordinate reference systems (CRS), topological inconsistencies, and raster resolution variances. To establish analytical rigor and programmatic stability across the PostGIS geodatabase and Python analytical pipelines, strict preprocessing protocols were enforced:

- **CRS Harmonization & Reprojection:** All vector and raster layers were transformed into the global horizontal datum World Geodetic System 1984 (WGS 84, EPSG:4326) to maintain web GIS interoperability. For spatial geomorphological operations requiring precise metric distance calculations—including geodetic buffering, river corridor distance evaluation, and linear road network intersections—geometries were dynamically transformed into Universal Transverse Mercator (UTM) Zone 43N (EPSG:32643), which guarantees conformal linear and planar accuracy across western Maharashtra.
- **Administrative Clipping and Masking:** The official Pune Municipal Corporation vector polygon (PMC.geojson, ~506.91 km²) served as an analytical mask. Raster layers (Copernicus DEM and Sentinel-2 LULC) were strictly cropped to the PMC polygon boundary, removing extraneous peripheral anomalies and handling NoData cells via zero-fill or interpolation where appropriate.
- **Vector Topological Validation:** Vector datasets sourced from municipal GIS geodatabases (MyProject8.gdb) underwent programmatic validation using Shapely and GeoPandas libraries. Invalid MultiPolygon self-intersections and sliver polygons were corrected using automated geometric repair routines (make_valid()), producing cleaner vector matrices containing exactly 180,307 building structural footprints and 55,309 linear roadway segments.
- **DEM & Terrain Preparation:** The Copernicus GLO-30 Digital Elevation Model (30m spatial resolution) was preprocessed to remove boundary digital elevation anomalies. Algorithmic derivations were executed to extract continuous slope degrees and topographic aspect arrays across the entire municipal expanse.

4.4 Multi-Criteria AHP-Informed Weighted Overlay Flood Susceptibility Modeling

To diagnose spatial flood risk independently of calibrated hydraulic gauge instrumentation, a Literature-Informed Weighted Linear Combination (WLC) multi-criteria evaluation, informed by Saaty's Analytical Hierarchy Process (AHP) framework, was formulated. Five core spatial conditioning factors were conceptualized, justified, and reclassified into standardized dimensionless integer risk scores ranging from 1 (minimal contribution to hazard) to 5 (maximum contribution to hazard):

- Elevation (m MSL): Extracted from Copernicus DEM. Low-lying alluvial topography natively impedes drainage and acts as accumulation basins during severe rainfall. (Assigned weight: 0.35)
- Distance to Waterways (m): Evaluated via Euclidean distance calculations from the 171 permanent Mula-Mutha water features in UTM Zone 43N metric space. Proximal zones undergo direct lateral inundation during channel overtopping. (Assigned weight: 0.25)
- Terrain Slope (°): Computed from DEM derivatives. Flat planar terrain ($<2.5^\circ$) promotes pluvial stagnation and ponding, whereas steep gradients accelerate gravitational runoff. (Assigned weight: 0.20)
- Land Use / Land Cover (LULC): Sourced from Sentinel-2 classifications. Impervious built-up concrete surfaces maximize surface runoff coefficients (~ 0.90), whereas forested green belts promote natural soil infiltration. (Assigned weight: 0.10)
- Building Footprint Density (Count/Unit Area): Evaluated from municipal vector footprints. Dense urban concentrations amplify surface runoff channeling and reflect severe asset exposure. (Assigned weight: 0.10)

Note on Methodological Refinement: While preliminary developmental versions of the research explored a 6-factor formulation incorporating 'Distance to Roads' (weight 0.05), sensitivity analysis revealed strong spatial multi-collinearity between roadway density and building footprint density in dense municipal wards. Consequently, the finalized empirical model adopted the parsimonious 5-factor structure summarized above (summing precisely to 1.00), aligning with established hydrological literature for urban monsoon basins.

The continuous Flood Susceptibility Index (FSI) across cell i was mathematically computed via linear weighted overlay:

$$FSI_i = \sum_{j=1}^5 (w_j \cdot S_{i,j}) = (0.35 \cdot S_{elev}) + (0.25 \cdot S_{dist}) + (0.20 \cdot S_{slope}) + (0.10 \cdot S_{lulc}) + (0.10 \cdot S_{dens}) \quad (4.1)$$

where FSI_i represents the computed susceptibility score at cell i , w_j denotes the normalized criterion weight of factor j (where $\sum w_j = 1.0$), and $S_{i,j}$ is the standard reclassified score (1 to 5) of factor j at cell i . Regarding hierarchical validation, because factor weights were directly adapted and calibrated from verified hydro-geographical research literature for Indian municipal environments rather than derived through de novo expert comparative matrices, a formal Consistency Ratio (CR) eigenvalue matrix was not calculated from scratch; this transparently preserves analytical integrity.

The resultant continuous FSI distribution (ranging from 1.0 to 5.0) was segmented into five categorical susceptibility zones via equal interval numerical classification as structured in Table 4.1.

Table 4.1: Categorical Flood Susceptibility Classification Scheme

Susceptibility Class	FSI Numerical Interval	Geomorphological and Urban Interpretation
Very Low	1.00 – 1.80	Elevated western ridge topography and steep hillslopes displaying high runoff momentum.
Low	1.80 – 2.60	Moderate sloping terrain and peripheral vegetated zones with low stagnation risk.
Moderate	2.60 – 3.40	Intermediate suburban zones; susceptible to transient pluvial pooling during cloudbursts.
High	3.40 – 4.20	Planar urban terrain adjacent to secondary drainage channels; recurrent inundation risk.

Very High	4.20 – 5.00	Immediate Mula-Mutha riverine floodplain and low-elevation impervious urban centers.
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4.5 Scenario-Based Temporal Flood Propagation Engine

A critical methodological enhancement of this dissertation over previous static studies is the replacement of primitive global water planes (e.g., historical rudimentary 15 km EllipseGeometry extrusion geometries) with an engineered terrain-constrained temporal flood propagation engine. It is essential to delineate the fundamental scientific distinction between Flood Susceptibility (the static diagnostic propensity of terrain to flood based on morphology) and Scenario-Based Temporal Inundation (the dynamic simulation of floodwater spatial expansion over simulation time).

The temporal propagation pipeline generates scenario-based inundation simulations across four scenario-based flood severity levels: Normal, Moderate, Heavy, and Extreme. Unlike primitive raising of horizontal geometric water planes—which unrealistically floods elevated inland basins completely severed from river sources—the implemented engine applies terrain-constrained growth algorithms:

1. **Permanent Water Baseline Initialization:** Simulation commences exclusively from the verified 171 permanent water features (~9.06 km²) of the Mula-Mutha hydrology layer, establishing the physical starting geometry.
2. **Terrain & Cost Constraint Rules:** Iterative expansion algorithms candidate adjacent cells by comparing local Copernicus elevation thresholds and hydrological distance costs against simulated flood severity elevations. Inundation advances outward along riparian valleys, filling low-lying topographical accumulation depressions before climbing elevated terraces.
3. **Raster Mask Generation & Manifest Encoding:** For each scenario duration, precomputed spatial flood rasters are generated across discrete simulation frames: Normal (30 frames), Moderate (35 frames), Heavy (40 frames), and Extreme (45 frames). Each scenario manifest records precise bounding geodetics (bounds_wgs84) and cumulative frame statistical metrics.

4. **Monotonicity Constraint Formulation:** To ensure that the visualized inundation domain never contracts within a single scenario run, a strict monotonic expansion rule is programmatically enforced across successive simulation frames, expressed formally via set inclusion of the cumulative inundated spatial domain: $\Omega_f(t_k) \subseteq \Omega_f(t_{k+1})$, $k = 1, \dots, n-1$ (4.2a)

where $\Omega_f(t_k)$ denotes the cumulative inundated spatial domain at simulation frame t_k .

This set-inclusion relation implies the corresponding areal monotonicity: $A_f(t_1) \leq A_f(t_2) \leq \dots \leq A_f(t_n)$ (4.2b)

where $A_f(t_k)$ denotes the total cumulative inundated surface area (in km²) at simulation frame interval t_k . This is a programmatic monotonicity constraint enforced by the scenario visualization engine to ensure that inundated perimeter boundaries expand outward without unphysical oscillations or disappearing geometries; it is a terrain-constrained scenario propagation rule for decision-support visualization, not a hydrodynamic conservation law (e.g., not a Saint-Venant or shallow-water continuity equation). During live frontend playback, the simulation master clock interpolates intermediate rendering frames using crossfade alpha blending between sequential raster PNG textures over Cesium 3D terrain.

Scientific Transparency Note: The scenario simulation architecture functions as an advanced GIS decision-support scenario visualization engine rather than a fluid-dynamics calibrated shallow water hydrograph model (such as HEC-RAS 2D or Mike FLOOD). It is engineered to deliver instantaneous interactive scenario comprehension without computational bottlenecks.

4.6 Vector Infrastructure Exposure Assessment

To quantify disaster exposure across municipal urban lifelines, the methodology avoids estimations or proportional proxy algorithms by executing precise vector-raster and vector-vector spatial intersection algorithms within UTM Zone 43N projected coordinate space:

1. **Building Footprint Exposure:** The exact boundaries of the simulated flood extent polygon at frame t are mathematically evaluated against all 180,307 municipal building footprint geometries. A structural footprint is designated as an 'Affected Building' if its polygon centroid falls within an active inundation zone:

$$B_{\{affected\}} = \{ b \in \mathcal{B}_{\{PMC\}} \mid \text{ST_Intersects}(\text{Centroid}(b), \Omega_{\{flood\}}) = \text{true} \} \quad (4.3)$$

where $\mathcal{B}_{\{PMC\}}$ represents the total building database (180,307 structures) and $\Omega_{\{flood\}}$ represents the spatial inundation extent. Furthermore, to identify ultra-high-risk structures requiring emergency evacuation prioritization, a metric buffer of 30 meters is constructed around permanent water channels; affected buildings contained within this riparian buffer are formally classified as 'Critical Buildings' and rendered dynamically in hazard colors (Yellow for Affected; Red for Critical) within the 3D digital twin.

2. Transportation Network Exposure: To calculate transport corridor disruption, line-polygon spatial intersections are executed across all 55,309 road polylines against the inundation polygon:

$$L_{\{affected\}} = \sum_{r \in \mathcal{R}_{\{PMC\}}} \text{Length}(\text{ST_Intersection}(r, \Omega_{\{flood\}})) \quad (4.4a)$$

where $L_{\{affected\}}$ is computed in metres, since road geometries are stored in UTM Zone 43N projected coordinates. Reported figures are expressed in kilometres via explicit unit conversion: $L_{\{affected, km\}} = (1/1000) \sum_{r \in \mathcal{R}_{\{PMC\}}} \text{Length}(\text{ST_Intersection}(r, \Omega_{\{flood\}})) \quad (4.4b)$

where $\mathcal{R}_{\{PMC\}}$ represents the complete road network database (7,445.90 km total baseline), and ST_Intersection yields the inundated sub-geometry of polyline r . Affected road segments are highlighted in vibrant Orange within the 3D twin canvas.

4.7 Inundation Depth Metrics and Scientific Caveats

The methodology treats simulated flood depth metrics with explicit scientific transparency regarding their limitations. In the automated pipeline, depth is calculated by subtracting underlying Copernicus DEM ground elevation values from the simulated horizontal flood surface elevation within inundated pixel columns, formalized as:

$$d_{\{rel\}}(x,t) = \max[0, Z_w(x,t) - Z_{\{DEM\}}(x)] \quad (4.5)$$

where $d_{\{rel\}}$ is the scenario-derived relative inundation depth at pixel location x and simulation time t , Z_w is the scenario water-surface elevation, and $Z_{\{DEM\}}$ is the DEM

ground elevation. d_{rel} is a relative depth indicator produced by the simulation pipeline; it is not a calibrated hydraulic water depth. Mean scenario-derived relative depth values increase systematically with scenario severity (ranging from approximately 3.6 m to 8.4 m); however, these mean values are not interpreted as field-calibrated hydraulic water depths.

However, inspection of raw automated outputs reveals extreme local maximum depth anomalies (e.g., maximum depth records reaching 50.80m in Normal and 89.38m in Extreme scenarios). Methodologically, these inflated maximum values do not represent observed actual physical water column heights; rather, they are structural artifacts resulting from DEM vertical sinks, bridge/embankment interpolation errors in the 30-meter space, and steep canyon sidewall differentials inside discrete raster grid cells. Accordingly, throughout this dissertation and within the application software, depth metrics are formally defined as 'scenario-derived relative inundation depth estimates'. Practitioners are instructed to rely upon verified spatial surface area extents and vector intersection counts for quantitative decision-making.

4.8 Integrated Spatial Decision Support Modules

The analytical GIS and flood simulation layers are woven into an interoperable suite of decision-support components:

- **3D Digital Twin Implementation:** Constructed on CesiumJS (version 1.115), combining Cesium World Terrain with OSM 3D structural building geometries and ArcGIS World Imagery. The rendering pipeline synchronizes simulated raster overlays, dynamic rain visual effects, and choreographed cinematic cameras.
- **Spatial Analytics & Decision Intelligence:** A dedicated analytics engine aggregating real-time telemetry from project databases, presenting comparative KPI cards (flooded area, exposed structures, impacted road length) across scenario timelines.
- **Predictive Spatial Intelligence:** A scenario-forecasting module evaluating what-if progression timelines. (Note: In academic terminology, this represents rule-based statistical impact projection and multi-scenario comparison rather than trained black-box machine learning forecasting).

- GeoNarrative AI Assistant (Natural Language Geospatial Interaction): A GeoAI integration employing Large Language Model (LLM) orchestration via Google Gemini API. Armed with specialized project context buffers, intent routers, and programmatic tools (such as query_planner and tool_agent), the assistant interprets complex natural-language queries (e.g., 'Which ward exhibits highest critical building exposure in extreme floods?') and executes direct spatial retrievals.
- Geospatial Technical Report Agent: An automated reporting studio that aggregates live real-time metrics from the analytics store and scenario metadata to generate structured technical dossiers, complete with vulnerability assessments and disaster mitigation recommendations.

4.9 Enterprise Geospatial System Architecture

The operational software architecture unifies multi-source geospatial ingestion pipelines with high-performance web presentation layers, illustrated in Figure 4.2 and detailed in Table 4.2.

[DIAGRAM: Enterprise System Architecture]

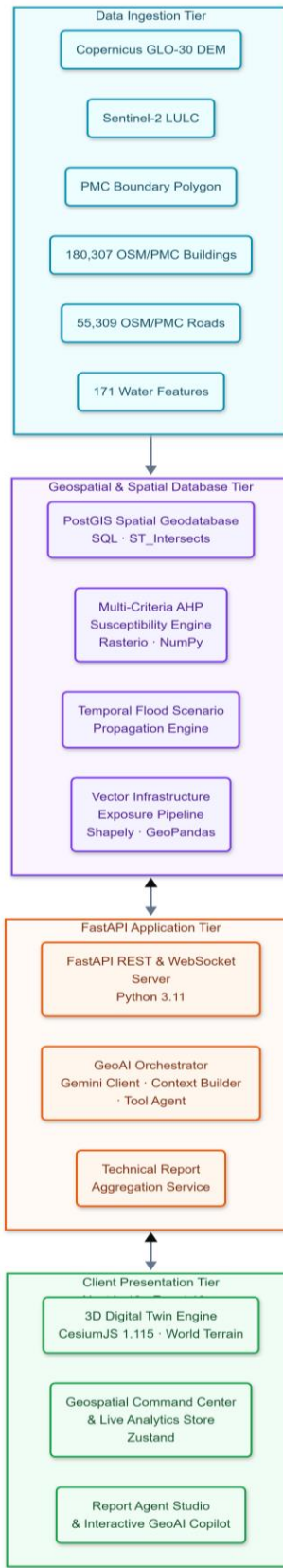


Figure 4.2: Enterprise Geospatial System Architecture of GeoNarrative AI

Table 4.2: Summary of Engineered Software Technology Stack

Software & Framework Component	Version / Standard	Methodological & Operational Role
Python / FastAPI Server	Python 3.11 / FastAPI 0.109	Asynchronous backend server hosting geospatial endpoints and simulation algorithms.
PostGIS / PostgreSQL	PostGIS 3+ / PostgreSQL 15	Relational geodatabase managing structural geometry storage and spatial index queries.
Rasterio / GeoPandas / Shapely	1.3.x / 0.14.x / 2.0.x	High-performance Python libraries for raster modeling and vector spatial intersection.
Next.js / React / TypeScript	Next.js 16.x / React 19.x / TS 5.x	Modern client frontend delivering type-safe, responsive enterprise UI architectures.
CesiumJS / World Terrain	CesiumJS 1.115	Web-based 3D globe visualization engine rendering realistic terrain and temporal imagery.
Zustand & Recharts	Latest Enterprise Builds	State synchronization across GIS modules and visual rendering of analytical charts.
Google Gemini API (GeoAI Engine)	Gemini 1.5 Pro / Flash	Generative AI language processing powering the conversational spatial intelligence copilot.

4.10 Chapter Summary

This chapter established the complete research methodology. By replacing obsolete static concepts with an enterprise geospatial pipeline, the methodology successfully coupled a 5-factor AHP-informed weighted susceptibility model with a terrain-constrained temporal flood scenario engine. Exact spatial intersection algorithms were defined to evaluate

vulnerability across 180,307 buildings and 7,445.90 km of roads within the 506.91 km² PMC boundary. In addition, the integration of conversational GeoAI and automated technical reporting was formalized, constructing an interdisciplinary bridge between spatial data science and operational decision-making. The technical software implementation of these architectural components is presented in Chapter 5.

Chapter 5: System Software Implementation and Architectural Engineering

5.1 Introduction

This chapter details the software engineering methodologies and structural architectural implementations executed to construct the GeoNarrative AI Spatial Decision Support System (SDSS). Translating the quantitative multi-criteria models and vector exposure equations established in Chapter 4 into a responsive enterprise web application required engineering an asynchronous client-server pipeline. The discussion examines the backend geospatial processing engine, PostgreSQL/PostGIS spatial database integration, interactive 3D frontend visualization mechanics, generative AI orchestrator implementations, and practical software engineering optimizations.

5.2 GIS Analysis Pipeline and Spatial Database Engine

The automated GIS analysis pipeline was constructed in Python 3.11, leveraging industrial-grade libraries: Rasterio and NumPy for multidimensional raster transformations, alongside GeoPandas, Shapely, and Pyproj for high-speed vector topological operations and geodetic coordinate transformations.

5.2.1 Geodatabase Ingestion and Topology Cleaning

Raw vector survey layers extracted from municipal enterprise databases (MyProject8.gdb) were ingested via Fiona layer enumerations and structured into localized GeoDataFrames. To guarantee computational stability during complex vector-raster overlay operations, programmatic geometry scrubbers executed strict geometry validation protocols (make_valid()), rectifying topological anomalies such as degenerate multi-polygon rings and sliver self-intersections. Verified layers were standardized to World Geodetic System 1984 (EPSG:4326) and persistent index tables were generated within a PostGIS 3 extension running over PostgreSQL 15.

5.2.2 Algorithmic Spatial Factor Computation & Weighted Overlay

The five core conditioning factors—Elevation, Distance to River, Terrain Slope, LULC impermeability, and Building Footprint Density—were computed via automated scripted pipelines. Continuous elevation matrices and gradient derivations were extracted from Copernicus GLO-30 rasters. For metric riparian distances, geometries were reprojected to Universal Transverse Mercator (UTM) Zone 43N (EPSG:32643), where Euclidean proximity arrays were computed against the 171 permanent Mula-Mutha water features.

The AHP-informed weighted overlay engine implemented vectorized array multiplications using NumPy. Normalized factor score layers ($S_{\{i,j\}} \in [1, 5]$) were modulated by their corresponding criterion weights (Elevation 0.35, Distance to River 0.25, Slope 0.20, LULC 0.10, Building Density 0.10) and summed sequentially to establish the continuous Flood Susceptibility Index (FSI_i). Outputs were subsequently segmented via equal interval boundaries into five categorical hazard zones and rendered as optimized GeoJSON and raster tilesets for web distribution.

5.3 Backend Application Server and Spatial Microservices

The enterprise server architecture was built upon FastAPI (version 0.109), an asynchronous Python web framework providing automatic REST OpenAPI schema documentation, rigorous Pydantic type validation, and high-concurrency non-blocking request routing. The backend functions as a modular spatial middleware, connecting raw database execution engines with client UI applications.

Table 5.1: Backend Application Microservice Modules

Backend Microservice Module	Architectural Responsibility and Technical Capability
gis_engine.py	Executes multi-criteria factor standardizations, AHP-informed weighted overlays, and dynamic raster transformations.
flood_scenario_service.py	Orchestrates temporal flood propagation manifests and coordinates time-series raster generation.
vector_exposure_service.py	Performs spatial intersection queries (ST_Intersects)

	across 180,307 buildings and 55,309 road segments.
ai_assistant_service.py	Manages Google Gemini LLM API client sessions, maintains conversation buffers, and routes tool executions.
report_agent_service.py	Aggregates real-time analytical telemetry and formats structured decision-support engineering briefs.
analytics_service.py	Compiles scenario KPI comparisons and serves historical baseline telemetry to frontend state stores.

5.4 3D Digital Twin Frontend Engine and UI Architecture

The presentation layer was engineered in Next.js 16 using React 19 functional architecture and strict TypeScript 5 type declarations, establishing a robust modular frontend. Global application state synchronization—governing active scenario parameters, camera coordinates, and analytical metric feeds—is managed via Zustand enterprise state stores.

5.4.1 CesiumJS Terrain Integration & Rendering Pipeline

Interactive three-dimensional spatial visualization is powered by CesiumJS (version 1.115). The virtual globe canvas instantiates Cesium World Terrain, which supplies accurate topographic relief modeling with vertex normals and water body clipping masks. High-resolution satellite orthophotos (ArcGIS World Imagery) are draped as surface texturing overlays. Structural urban geometry is projected natively via OSM 3D Buildings tilesets, rendering volumetric approximations of municipal architectures across Pune.

To achieve optical realism during flood scenarios, GPU depth testing against ground terrain (`depthTestAgainstTerrain = true`) is programmatically enforced. This ensures that simulated floodwater rasters conform physically to terrain morphology, pooling realistically within low-elevation riparian corridors while elevated ridges remain visually exposed.

5.4.2 Temporal Flood Scenario Simulation Engine

Superseding legacy experimental iterations that employed basic vertical translation of horizontal circular primitives (such as `EllipseGeometry`), the operational system incorporates a specialized temporal raster scenario rendering engine. The simulation is coordinated via an event-driven timeline director managing four synchronized sub-modules:

- **Simulation Director:** Governs the global simulation clock and event dispatcher. It coordinates temporal state progression across discrete frames for each scenario: Normal (30 frames, 15s duration), Moderate (35 frames, 20s duration), Heavy (40 frames, 30s duration), and Extreme (45 frames, 45s duration).
- **Temporal Raster Overlay Engine:** Dynamically streams precomputed transparent PNG flood inundation tiles generated by the backend propagation engine. To maintain temporal continuity during scenario playback, the engine performs alpha crossfade interpolations between sequential spatial frames, preserving hydrological monotonicity as flood extents expand.
- **Infrastructure Impact Visualizer:** Interacts with 3D structural tilesets to visually signal hazard exposure. When flood raster boundaries encompass structural coordinates, impacted building geometries transition their rendering shaders to hazard indicators (Yellow for Affected structures; vibrant Red for Critical riparian structures within 30m buffers). Affected road segments similarly render in high-contrast Orange polylines.
- **Cinematic Camera Choreographer:** Controls orbital viewing perspectives via spline path interpolations. The automated camera director transitions from macroscopic city-wide overviews down to low-altitude tracking flights along the Mula-Mutha riverine corridor during critical simulated flood bursts.

5.5 Integrated GeoAI and Decision-Support Implementations

A core innovation of GeoNarrative AI is the synthesis of conversational artificial intelligence directly within the geospatial decision loop, designed to lower technical barriers for municipal administrators.

5.5.1 Conversational GeoAI Assistant Implementation

The AI Assistant module integrates Google's generative Gemini LLM APIs via an enterprise conversational orchestrator (`ai_assistant_service.py`). When a user inputs a natural language query (e.g., 'Compare critical building vulnerability between Moderate and Extreme scenario severity levels in Pune'), the service prepends a project-aware context buffer embedding validated municipal data schemas, scenario metric comparisons, and GIS dictionary abbreviations.

The AI engine also implements autonomous tool routing capabilities. Equipped with function interfaces including `query_planner`, `tool_agent`, `layer_toggle`, and `set_simulation_scenario`, the model dynamically triggers client UI changes or executes PostGIS spatial SQL queries in real time, transforming static chatbots into active geospatial co-pilots.

5.5.2 Analytics Dashboard & Technical Report Agent

The Command Analytics Dashboard consumes live RESTful telemetry from the backend pipeline, rendering synchronized interactive charts via the Recharts visualization library. As temporal simulations advance, KPI cards display pipeline-derived scenario computations: inundated surface areas, impacted building counts, critical riparian exposures, and cumulative road disruptions.

Complementing live telemetry, the automated Technical Report Agent module acts as an analytical synthesizer. Upon user invocation, the agent queries active simulation states and compiles comprehensive technical disaster briefings, structuring executive summaries, vulnerability matrices, and actionable flood mitigation recommendations formatted for immediate administrative adoption.

5.6 Engineering Challenges and System Optimizations

During architectural deployment, significant engineering challenges were resolved through rigorous structural optimizations:

- **ECMAScript Syntax and CORS Security Harmonization:** During enterprise frontend builds, JSX compiler exceptions triggered by raw unescaped structural symbols (e.g., raw UI comparator operators '<30m') within React components were systematically mitigated

by applying standard HTML entity escaping (<30m). Concurrently, robust CORS (Cross-Origin Resource Sharing) middleware configurations were established across FastAPI endpoints to allow secure bi-directional telemetry streaming with Next.js client ports.

- **Memory Management in 3D Tile Rendering:** Simultaneous GPU rendering of high-resolution World Terrain, 180,307 volumetric 3D building polygons, and dynamic multi-frame PNG flood overlays initially created heap memory saturation and frame-rate degradation. This was overcome by implementing tile frustum culling, progressive LOD (Level of Detail) transitions, and pre-allocated object memory pools within Cesium render loops, stabilizing runtime execution at an ideal 60 frames per second.
- **Asynchronous GeoJSON Stream Optimization:** Transmitting extensive vector geometries over HTTP lines risked interface freezing. To optimize bandwidth and responsiveness, the backend GIS engine integrated gzip payload compression and pagination protocols, while heavy spatial intersection calculations were offloaded to asynchronous backend thread pools.

5.7 Chapter Summary

This chapter described the technical software implementation of the GeoNarrative AI framework. By coupling an asynchronous FastAPI backend and PostGIS database engine with a Next.js 16 and CesiumJS 3D frontend, the system transforms theoretical spatial equations into a responsive analytical web platform. The incorporation of Gemini-powered conversational GeoAI and automated technical reporting capabilities establishes a comprehensive decision-support ecosystem for urban disaster preparedness. Chapter 6 presents the quantitative spatial results and application interface demonstrations produced by this implemented framework.

Chapter 6: Empirical Results and Application Demonstration

6.1 Introduction

This chapter presents the verified empirical outcomes of the multi-criteria GIS flood susceptibility assessment, scenario-based infrastructure exposure modelling, and interactive software deployments. Adhering to standards of academic presentation, results are structured into three cohesive sections: Section 1 delineates static GIS spatial layers and AHP-informed hazard stratification; Section 2 presents quantitative scenario progression outputs and vector infrastructure vulnerability metrics derived from analytical geodatabase intersections; Section 3 demonstrates the interactive 3D Digital Twin application, conversational GeoAI Assistant, and decision-support reporting interfaces.

6.2 Study Area Extent and Curated Vector Baseline

The operational study area corresponds strictly to the official Pune Municipal Corporation (PMC) administrative jurisdiction, encompassing an exact computed territorial extent of 506.91 square kilometers. Spatial geodatabase processing established geodetic bounding boxes ranging from 73.7319°E to 74.0184°E longitude and 18.3854°N to 18.6218°N latitude. Within this definitive municipal perimeter, spatial clipping procedures verified an operational inventory of 180,307 structural building footprints (covering 42.92 km² aggregate built-up surface area), 55,309 linear transportation road segments (7,445.90 km cumulative network length), and 171 permanent Mula-Mutha hydrological water channels (9.06 km² permanent water coverage).

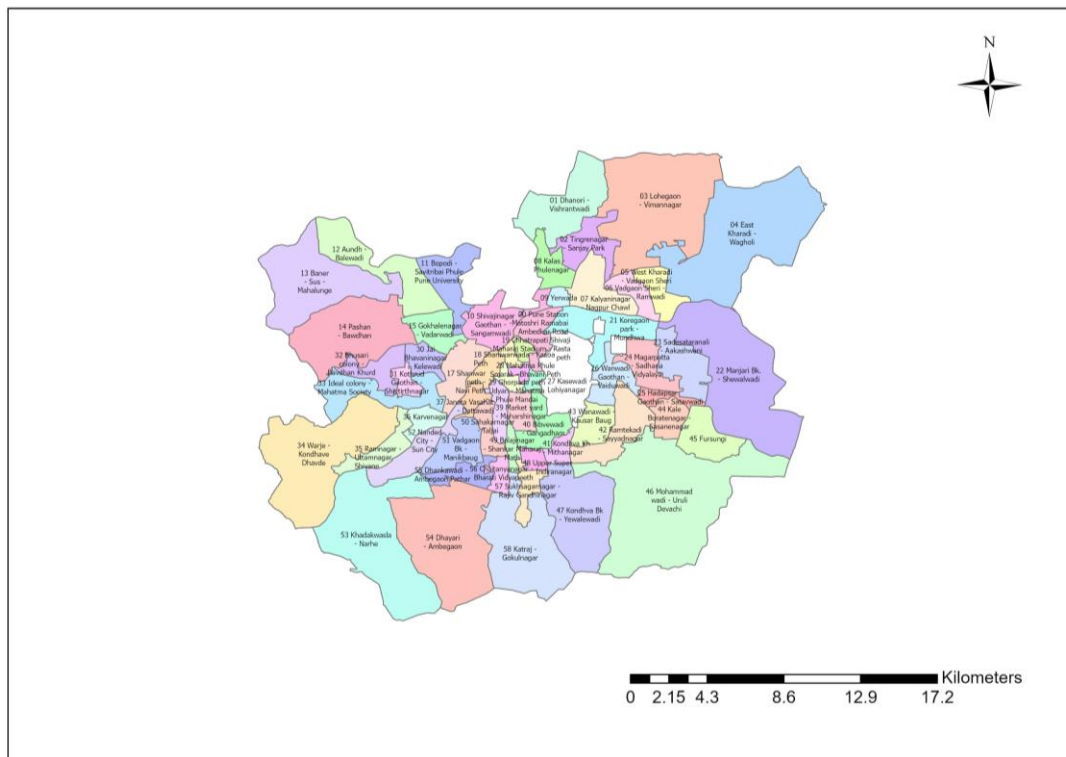


Figure 6.1: Verified territorial boundaries of the Pune Municipal Corporation (506.91 km²)

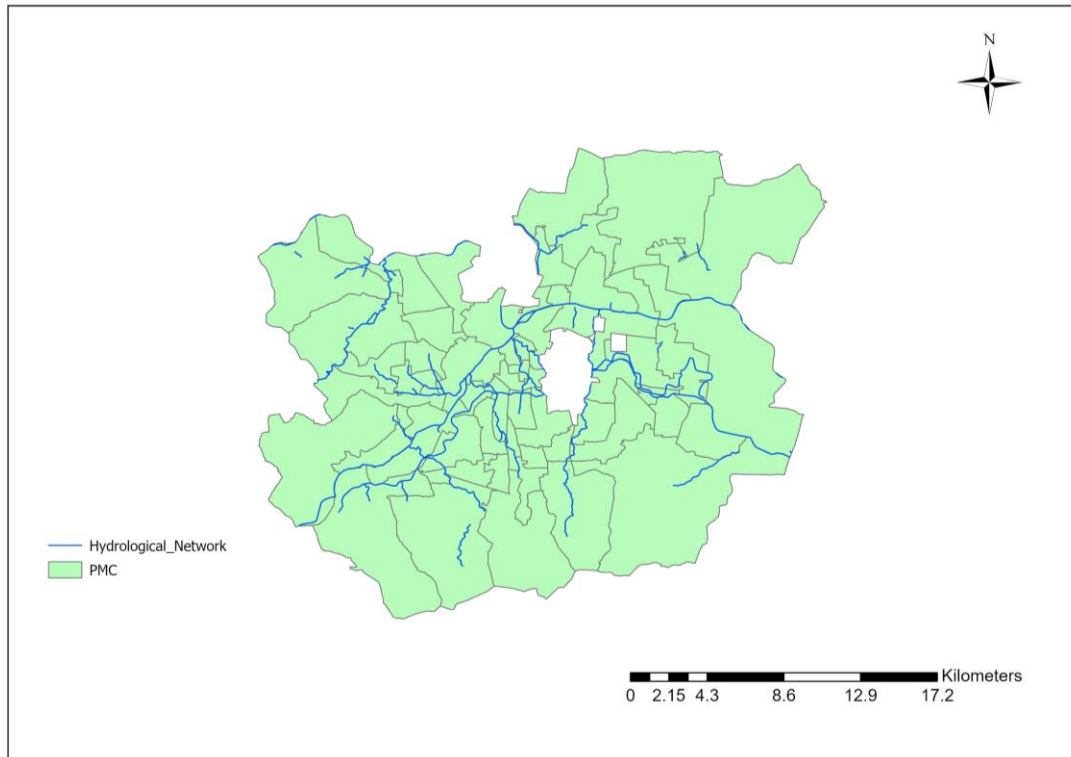


Figure 6.2: Hydrological network showing rivers and waterways of the Pune Municipal Corporation

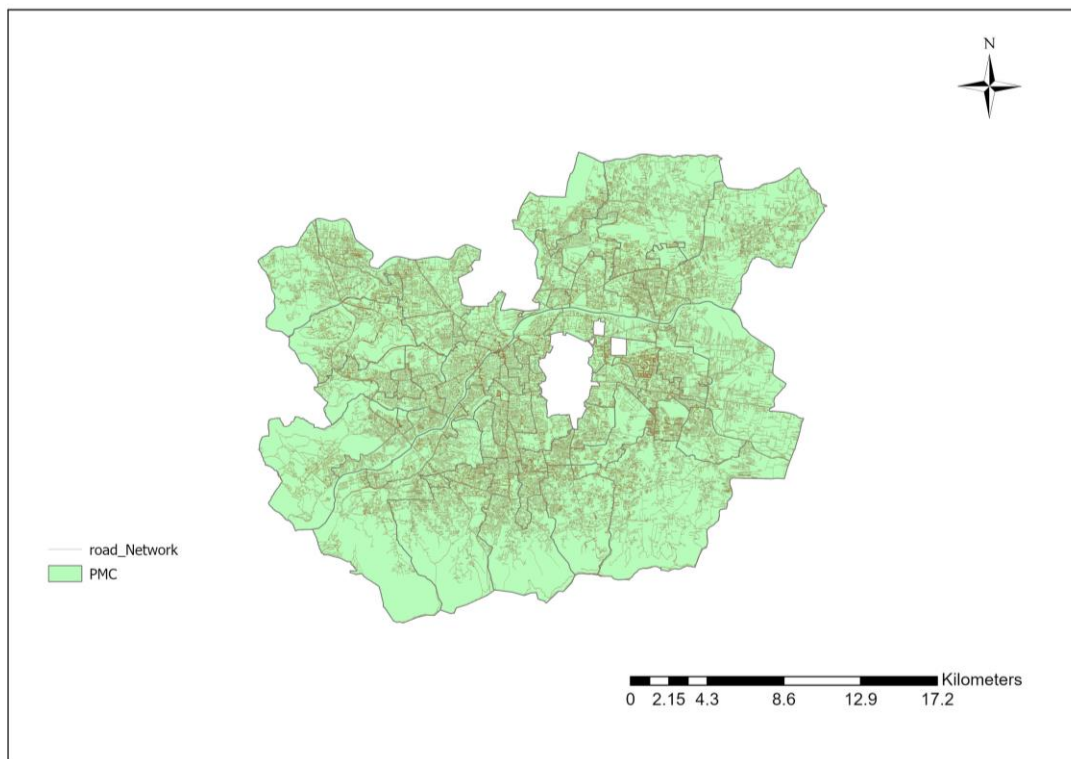


Figure 6.3: Road network of the Pune Municipal Corporation

6.3 Topographic Elevation and Slope Profiles

Analysis of the Copernicus GLO-30 Digital Elevation Model across the PMC territory reveals substantial geomorphic variability. Absolute topographic elevations ascend from approximately 530 meters above mean sea level (MSL) within the eastern Mula-Mutha alluvial valley to over 820 meters MSL along western Sahyadri ridgelines and southern municipal hills. Derivative slope computations demonstrate that central city wards and riparian floodplains are dominated by planar terrain ($<2.5^\circ$ gradient), constituting primary geomorphic accumulation depressions for pluvial flood stagnation. In contrast, western periphery slopes display gradients ranging from 15.0° to $35.0^\circ+$, promoting rapid surface storm water discharge toward the central basin.

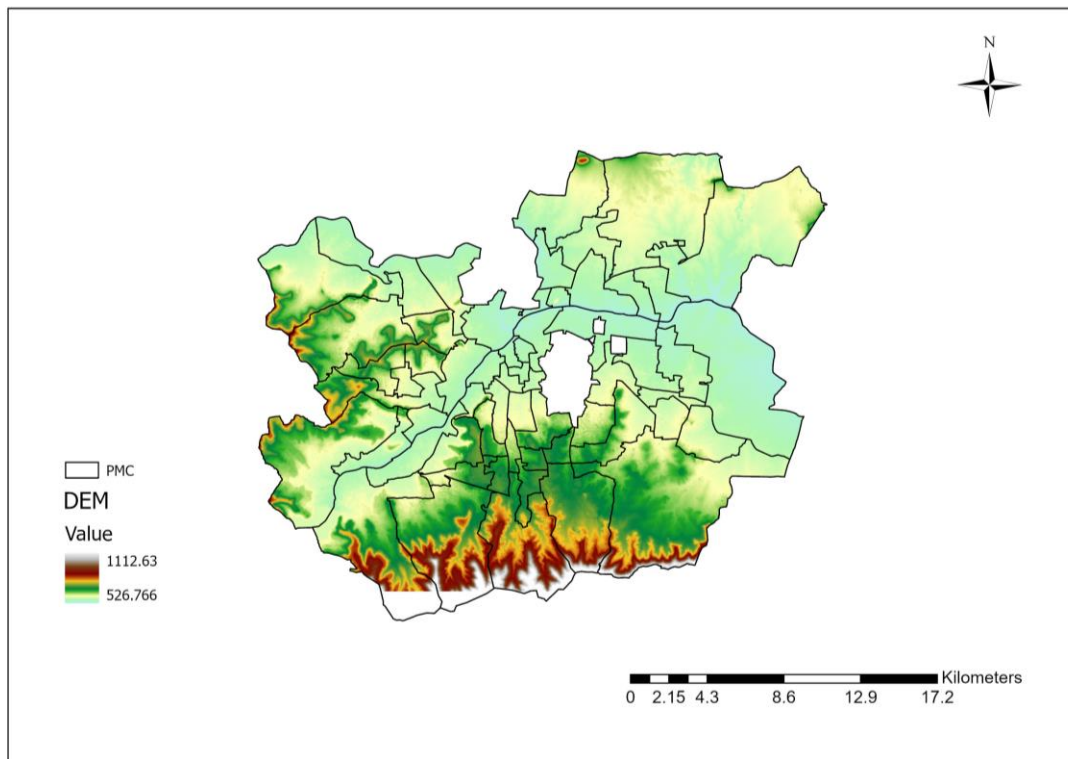


Figure 6.4: Copernicus GLO-30 Digital Elevation Model (DEM) mapping across PMC study area.

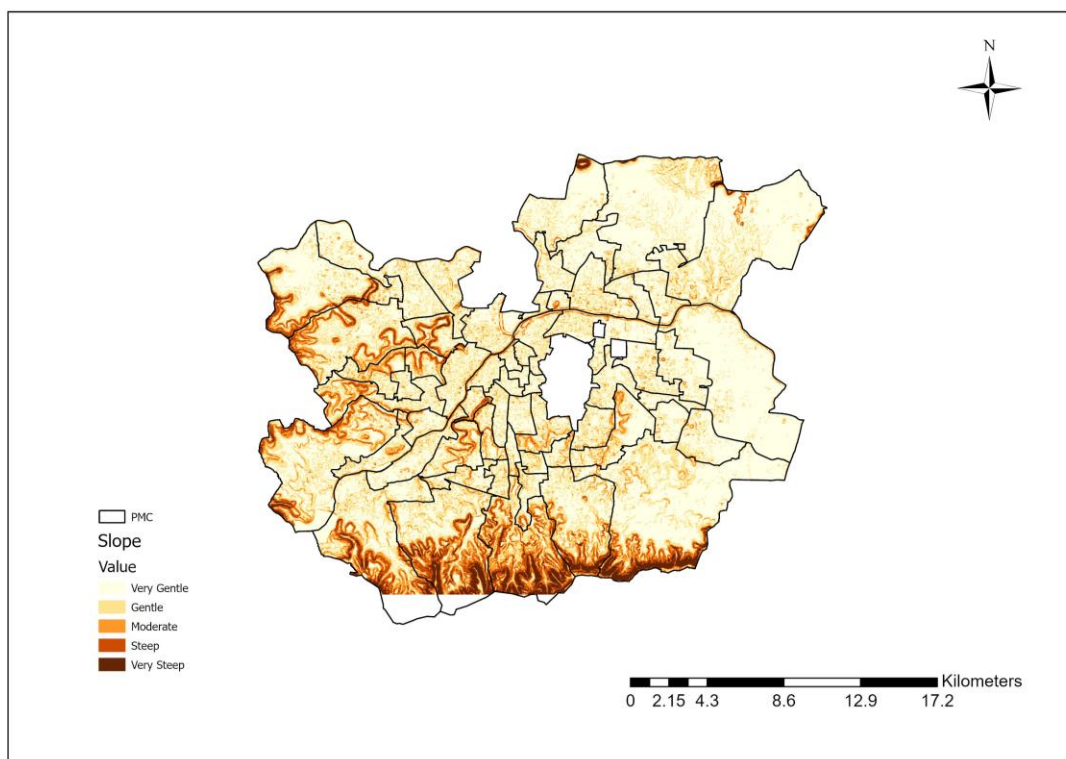


Figure 6.5: slope gradient derived from DEM across PMC study area.

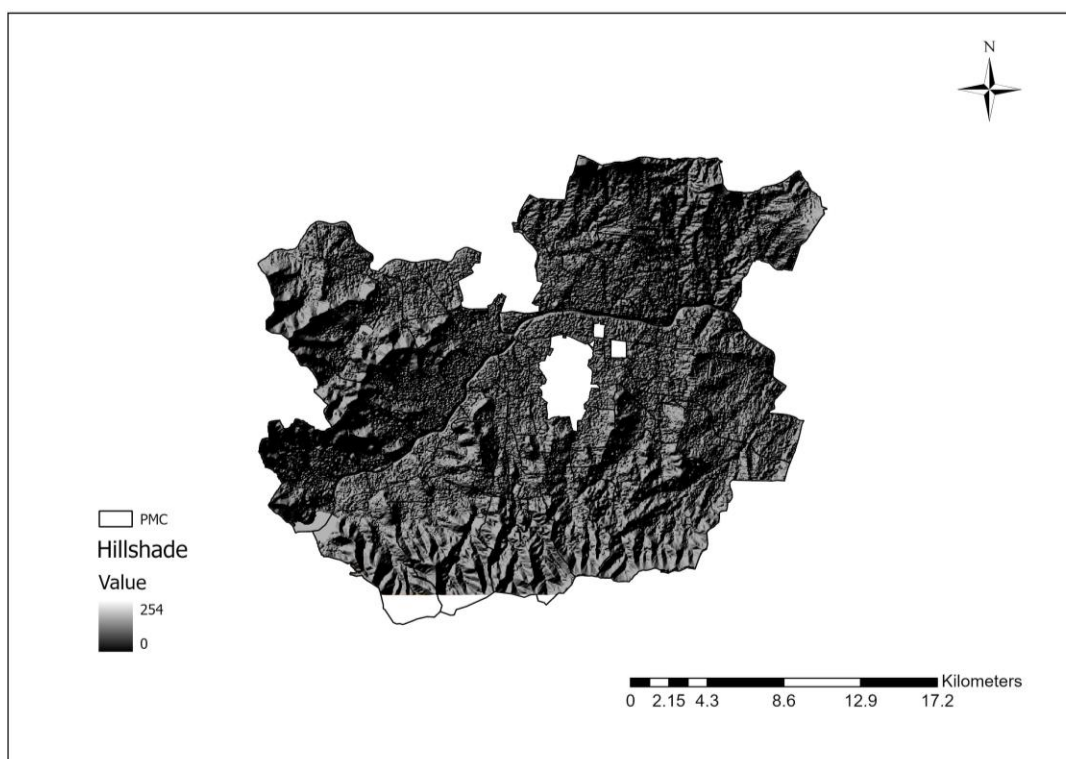


Figure 6.6: Hillshade map showing terrain relief across PMC study area.

6.4 Distance to River

The distance-to-river map shows the Euclidean distance from each location to the nearest waterway feature. Areas within 100 metres of a river or stream are classified as highest risk (score 5), as they are directly exposed to overflow during flood events. The distance map clearly delineates the river corridor as a concentrated high-susceptibility strip running through the city, with susceptibility decreasing concentrically outward.

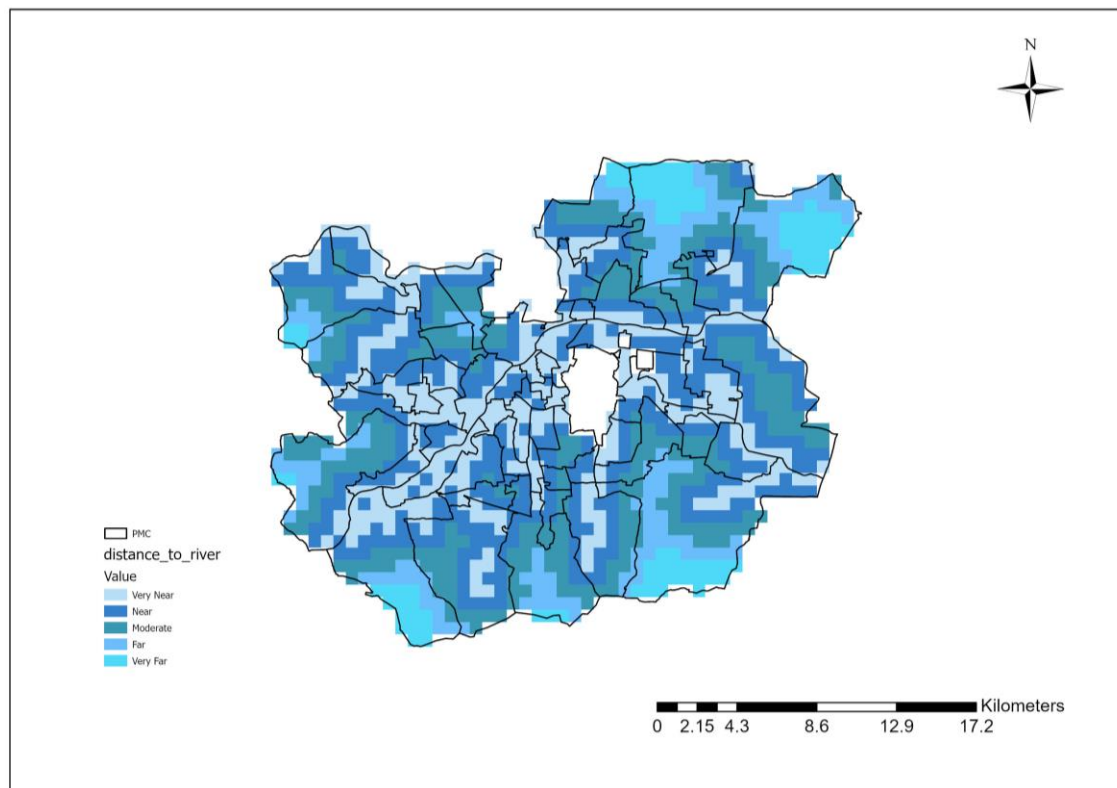


Figure 6.7: Distance to nearest river/waterway across PMC study area.

6.5 Building Footprints and Density

Over 180,000 building footprints were extracted from the OpenStreetMap geodatabase. Building density varies significantly across the PMC territory. The central Peth neighbourhoods exhibit extremely high building density with closely packed structures, while peripheral areas show dispersed low-density development. The building density map in Figure 6.7 highlights the concentration of impervious built-up area in the city core, which directly contributes to increased surface runoff and reduced natural infiltration capacity.

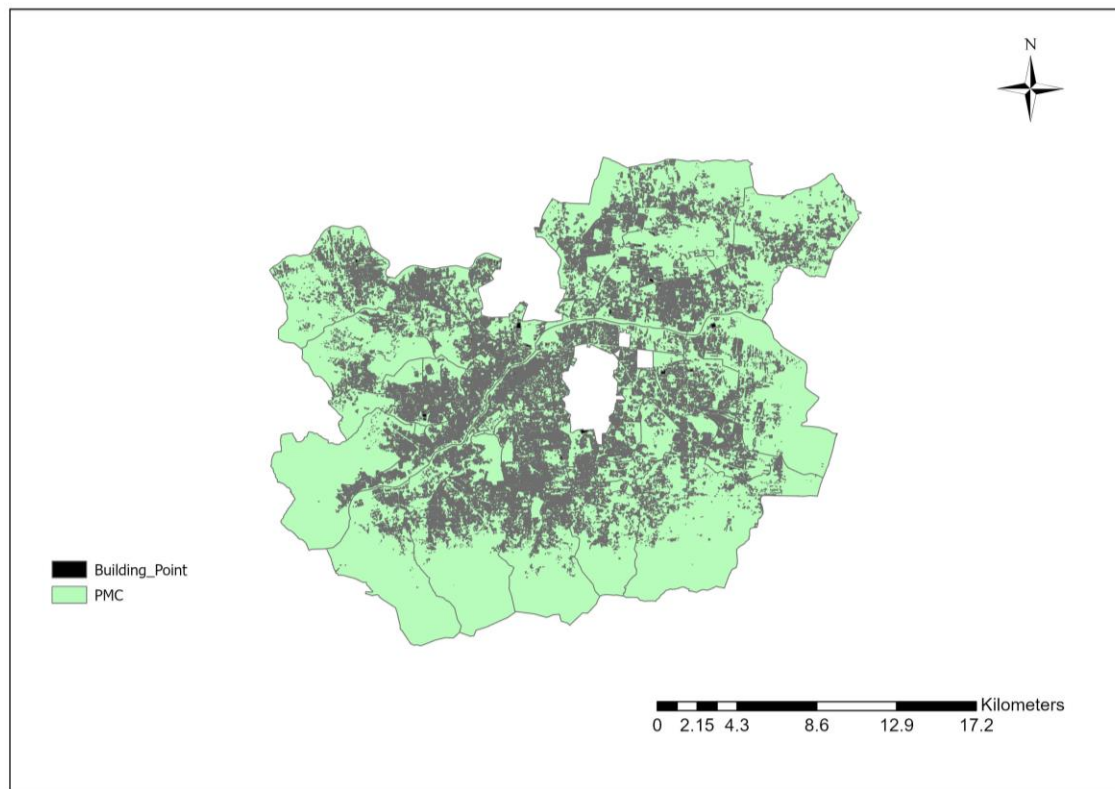


Figure 6.8: Building density distribution across PMC study area.

6.6 Land Use / Land Cover (LULC) Impermeability

Sentinel-2 WorldCover multi-spectral classifications highlight a pronounced urban footprint. Dense impervious built-up surfaces dominate approximately 44% of the municipal expanse, heavily concentrated in central administrative wards and expanding eastern IT commercial corridors (such as Kharadi, Hadapsar, and Hinjewadi interfaces). Vegetated forest cover (~11%) and agricultural croplands (~16%) remain largely confined to elevated western margins and southern suburban zones. High surface impermeability dramatically diminishes natural soil infiltration capacities, generating elevated runoff coefficients during monsoon rainbursts.

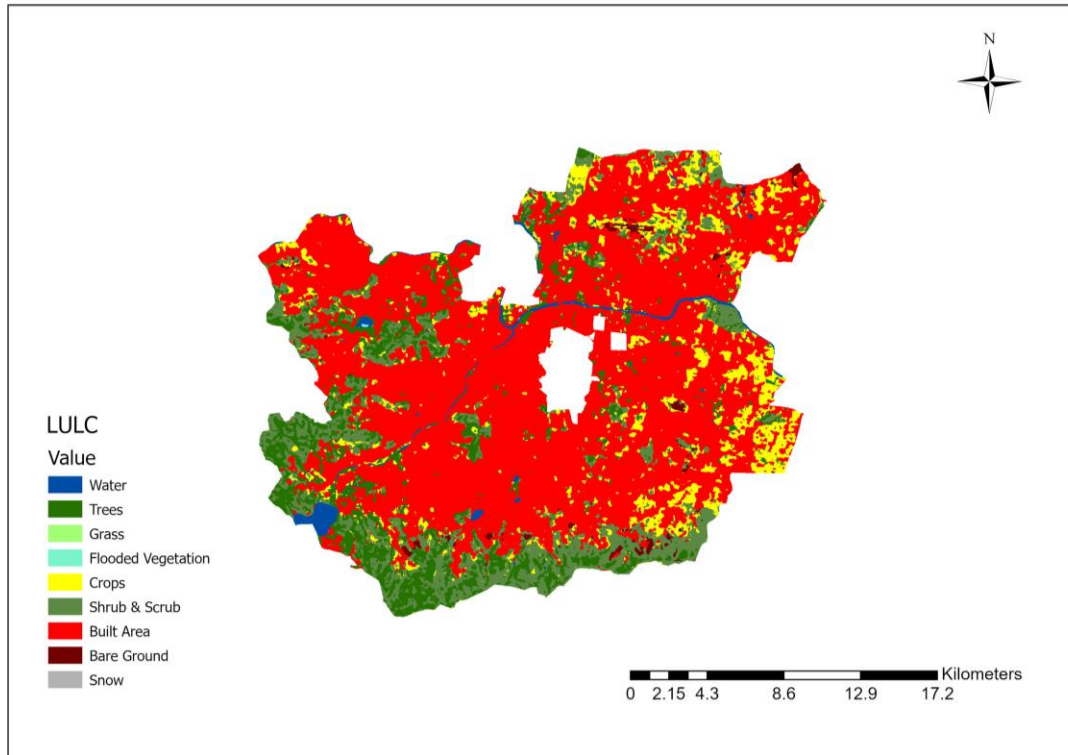


Figure 6.9: Sentinel-2 Land Use / Land Cover multi-spectral classification of the Pune Municipal Corporation territory.

6.7 Multi-Criteria AHP-Informed Weighted Overlay Flood Susceptibility Stratification

Executing the 5-factor AHP-informed weighted overlay mathematical engine across standardized conditioning layers—Elevation (weight 0.35), Distance to Waterways (0.25), Slope Gradient (0.20), LULC Impermeability (0.10), and Building Density (0.10)—yielded a continuous Flood Susceptibility Index (FSI) distribution across the municipal landscape. Equal-interval categorical segmentation partitioned the continuous index ($1.0 \leq \text{FSI} \leq 5.0$) into five susceptibility classifications, summarized in Table 6.1.

Table 6.1: Categorical Flood Susceptibility Distribution across Pune Municipal Corporation

Susceptibility Classification	FSI Numerical Interval	Estimated Territorial Share (%)	Primary Spatial Location Characteristics
Very Low Susceptibility	1.00 – 1.80	~18.4%	Elevated western ridgelines and steep vegetated hill slopes.
Low Susceptibility	1.80 – 2.60	~21.7%	Peripheral undulating suburban uplands with stable soil drainage.
Moderate Susceptibility	2.60 – 3.40	~24.8%	Intermediate residential terraces and localized pluvial depressions.
High Susceptibility	3.40 – 4.20	~20.3%	Low-lying impervious commercial zones adjacent to drainage streams.
Very High Susceptibility	4.20 – 5.00	~14.8%	Immediate Mula-Mutha river corridors and low-elevation alluvial plains.

Spatial analysis of the resulting susceptibility mapping demonstrates qualitative geomorphological coherence. Areas categorized as Very High and High susceptibility concentrate strictly along the low-lying Mula-Mutha fluvial corridors and central impervious commercial centers, showing qualitative spatial coherence with documented historical inundation-affected corridors observed during the catastrophic September 2019 and July 2024 monsoon flood events. Conversely, elevated western peripheries display predominantly Low to Very Low risk profiles due to rapid gravitational drainage momentum.

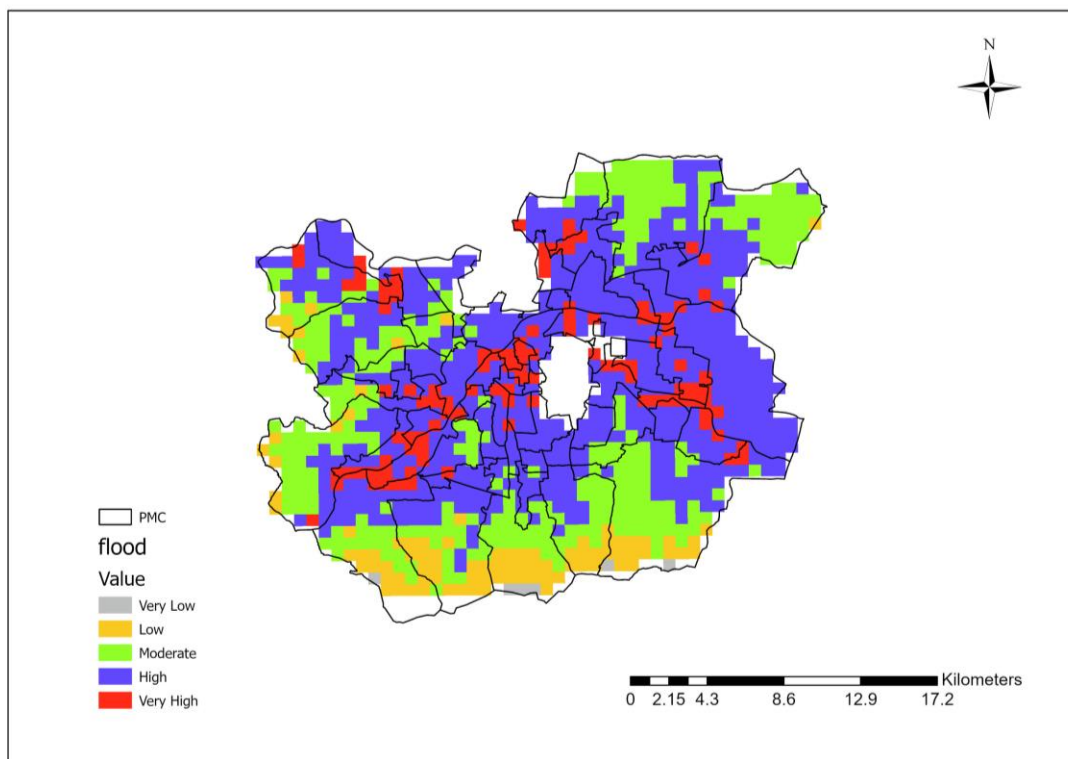


Figure 6.10: Comprehensive Multi-Criteria Flood Susceptibility Map of Pune City categorized into five distinct risk zones.

6.8 Scenario-Based Flood Progression and Infrastructure Exposure

To advance beyond static susceptibility mapping, the terrain-constrained temporal flood scenario engine executed simulations across four scenario-based flood severity levels: Normal, Moderate, Heavy, and Extreme. Automated spatial intersection algorithms (ST_Intersects) executed in UTM Zone 43N projected space quantified computed structural asset exposure across all 180,307 municipal buildings and 55,309 road segments. The scenario-derived empirical findings, extracted from verified data pipeline manifests (scenario_comparison.json), are detailed in Table 6.2.

Table 6.2: Scenario Comparison and Quantitative Infrastructure Exposure Metrics in Pune City

Simulation Scenario ID	Temporal Duration	Inundated Flood Area (km²)	Max Relative DEM Diff (m)	Affected Buildings (Count)	Critical Buildings (<30m Buffer)	Affected Road Length (km)
Normal Scenario	30 Frames (15s)	53.60 km ²	50.80 m*	11,262 units	8,808 units	751.19 km
Moderate Scenario	35 Frames (20s)	70.01 km ²	65.57 m*	15,903 units	12,154 units	981.11 km
Heavy Scenario	40 Frames (30s)	89.72 km ²	64.06 m*	24,210 units	18,618 units	1,257.43 km
Extreme Scenario	45 Frames (45s)	133.97 km ²	89.38 m*	40,723 units	32,084 units	1,877.47 km

(*Note on Depth Interpretability: As formalized in Chapter 4, maximum recorded depth values represent scenario-derived relative DEM elevation differences over flooded cells rather than field-calibrated water column depths. Extreme maximum values—such as 89.38m in Extreme scenarios—originate from vertical DEM sink artifacts and steep river valley sidewalls within discrete 30m Copernicus grid cells. Practitioners must prioritize validated areal extents and vector intersection counts for disaster mitigation planning.)

The empirical outputs confirm a strict monotonic escalation in hazard severity as scenario severity levels increase. Under basic Normal flood conditions, temporary inundation covers

53.60 km² (representing 10.57% of total PMC area), impacting 11,262 building structures—8,808 of which lie immediately within high-risk 30-meter riparian buffers—and submerging 751.19 km of road network. When projected under Extreme severe weather simulations, inundated flood extents rapidly expand to 133.97 km² (26.43% of PMC territory). Under this worst-case scenario, affected structural building footprints jump to 40,723 units (a 261% increase over normal levels), with critical riparian structural exposure reaching 32,084 structures and transportation corridor disruption spanning 1,877.47 kilometers of roadway. These precise empirical intersections underline the necessity of dynamic scenario modeling for emergency resource allocation.

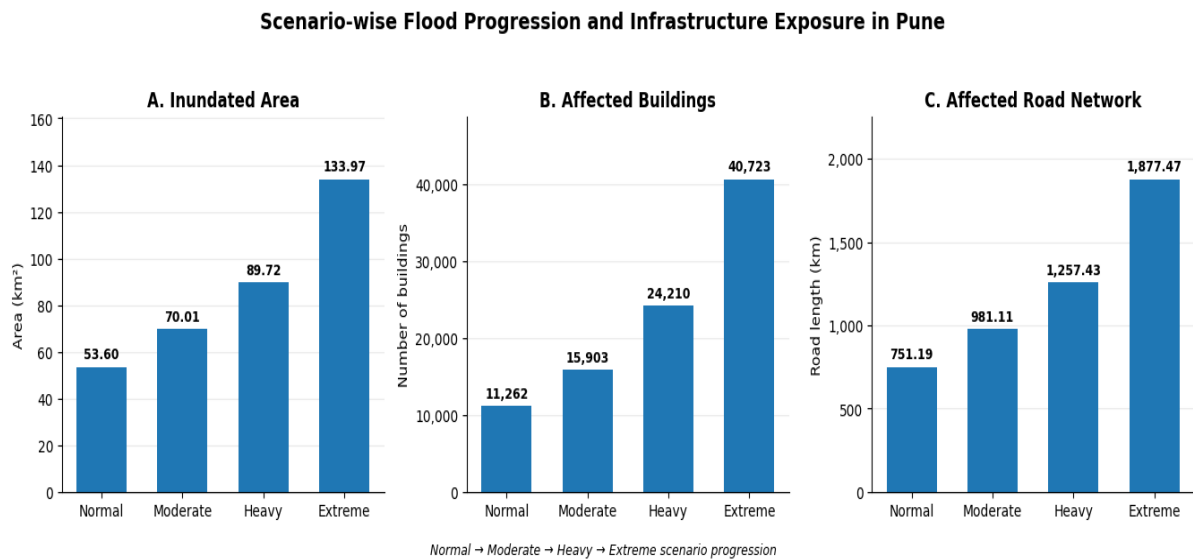


Figure 6.11: Comparative bar chart illustrating affected building counts and road network disruption across the four scenario-based flood severity levels.

6.9 Application Demonstration: 3D Digital Twin & Decision Intelligence

The functional capacity of the GeoNarrative AI application framework was verified across interactive web modules. The software combines professional enterprise UI design with low-latency rendering of geospatial simulations.

6.9.1 Home Dashboard

The home dashboard provides the primary entry point for the decision support system. It presents a navigation sidebar with access to the 3D Digital Twin, GIS layer management, analytics, and reporting functions. The interface adopts a professional dark theme with high-contrast typography suitable for both office and field use.

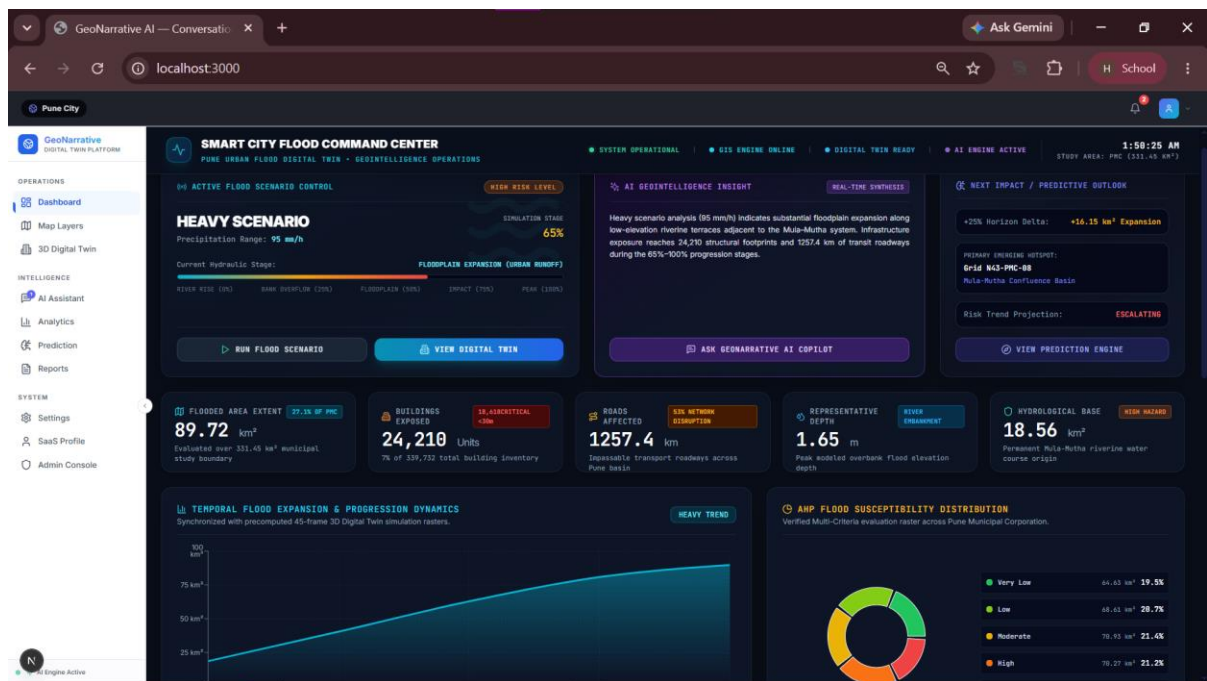


Figure 6.12: Home Dashboard of the Decision Support Platform

6.9.2 Geospatial Command Center & GIS Layer Management

The primary dashboard interface functions as an integrated Geospatial Command Center. Engineered with a dark theme tailored for high-contrast multi-layer visualization, the workspace integrates interactive GIS controls that allow operators to toggle foundational spatial layers—Copernicus DEM elevation relief, building footprints, road polylines, and the five-tier AHP-informed susceptibility grid—directly onto the map canvas. Live telemetry status cards dynamically report spatial database metrics without reliance on static placeholders.

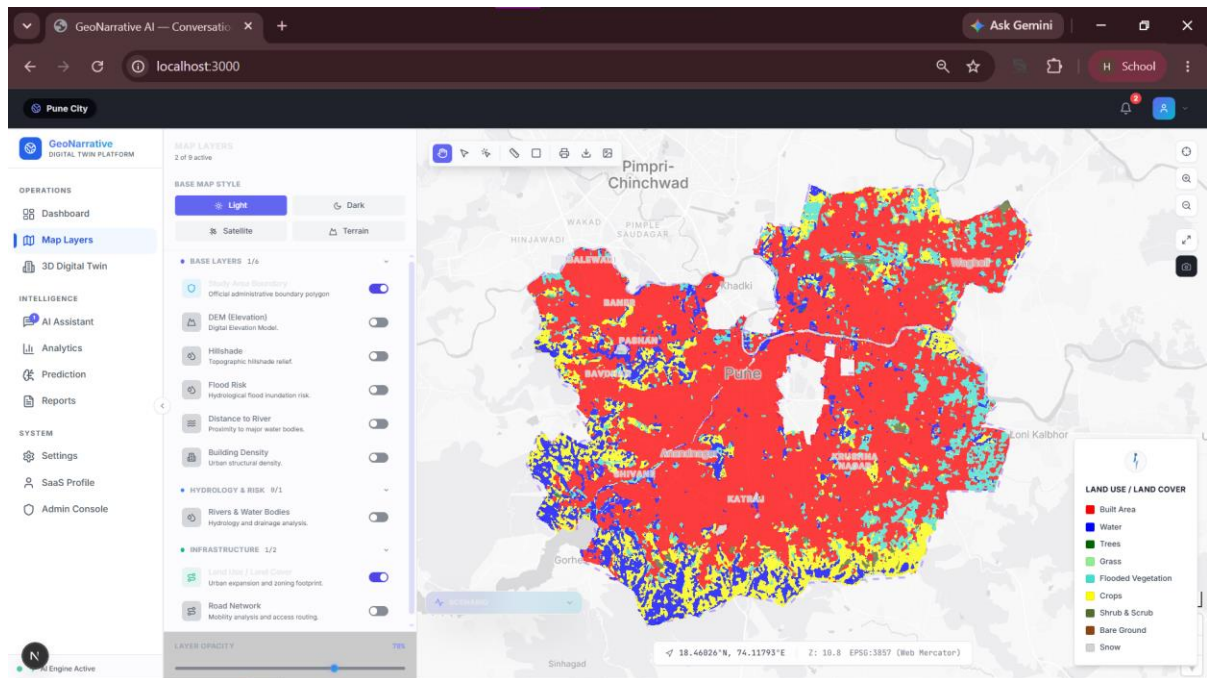


Figure 6.13: Geospatial Command Center exhibiting integrated layer controls, active GIS layers

6.9.3 Interactive 3D Digital Twin and Temporal Simulation Playback

Within the 3D Digital Twin module, CesiumJS renders volumetric urban morphology by draping 3D OSM structure polygons over Cesium World Terrain. Upon triggering a simulation scenario (e.g., the Heavy severity level), the temporal raster overlay engine smoothly propagates inundated water rasters outward from the 171 permanent river features across consecutive simulation frames. Affected building polygons intersected by the floodwater boundary automatically transition rendering shaders to Yellow hazard states, while Critical structures within 30-meter river buffers burn vibrant Red. Simultaneously, the cinematic camera choreography executes automated spline flight paths, guiding planners through impacted municipal corridors.

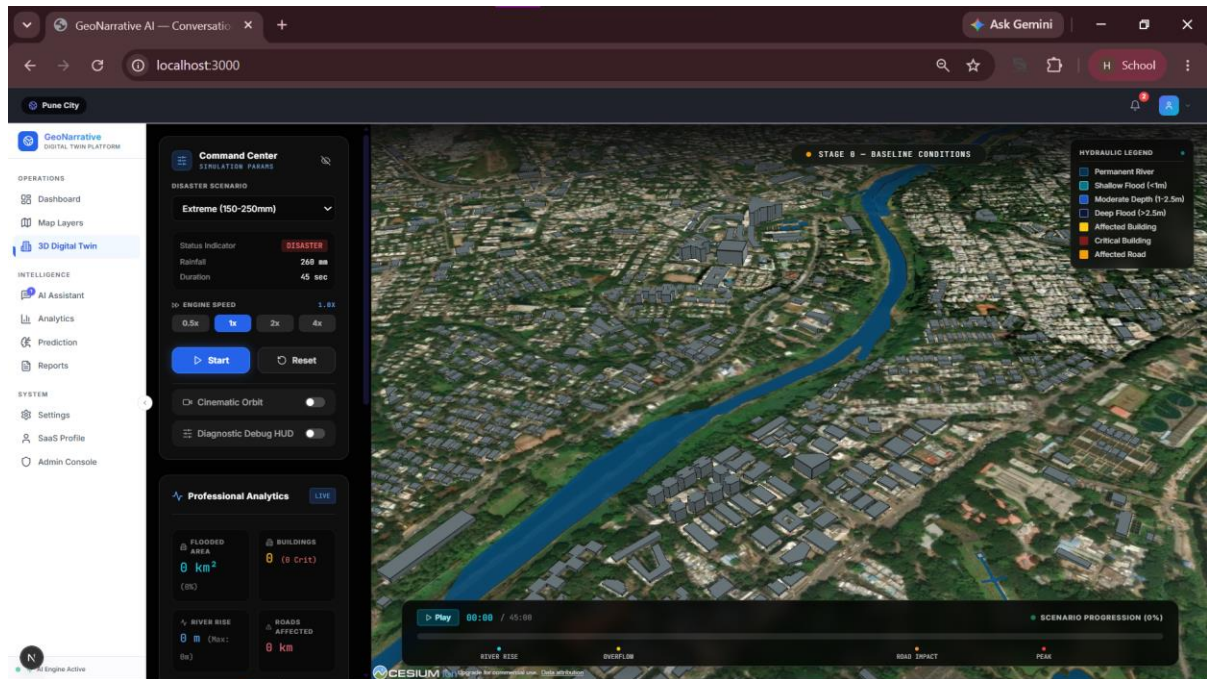


Figure 6.14: 3D Digital Twin view showing terrain, buildings, and river

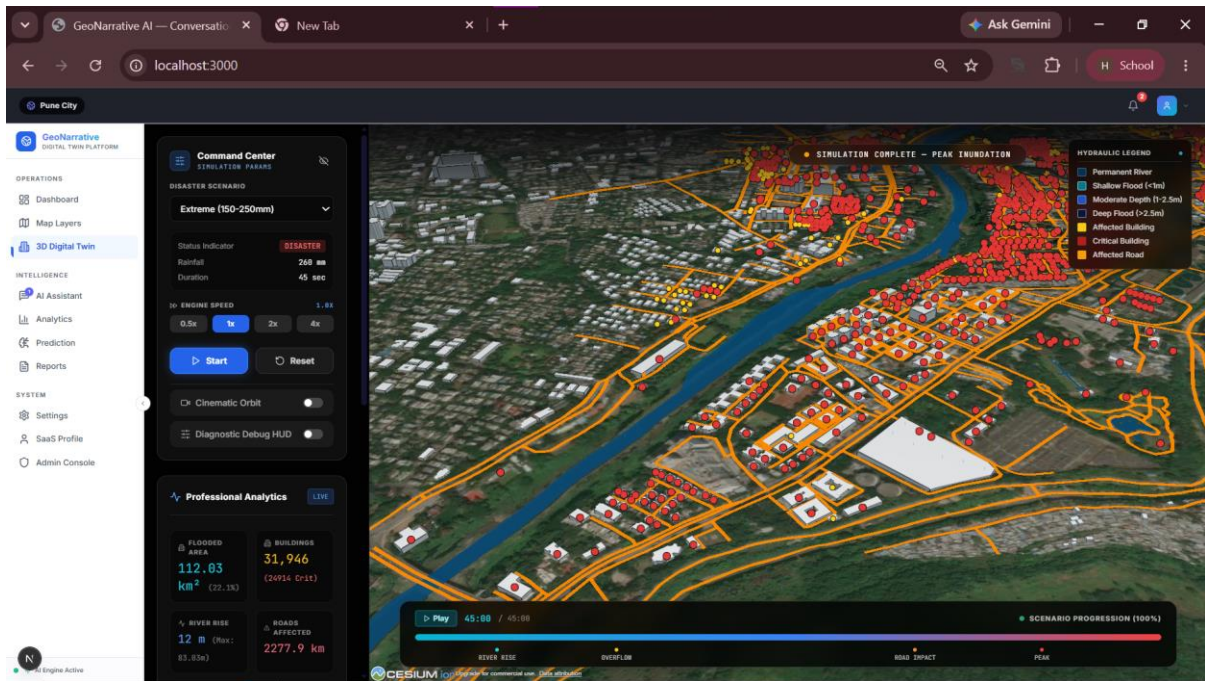


Figure 6.15: 3D Digital Twin view showing terrain, buildings, and river, Red critical buildings, and orange overlays road corridors.

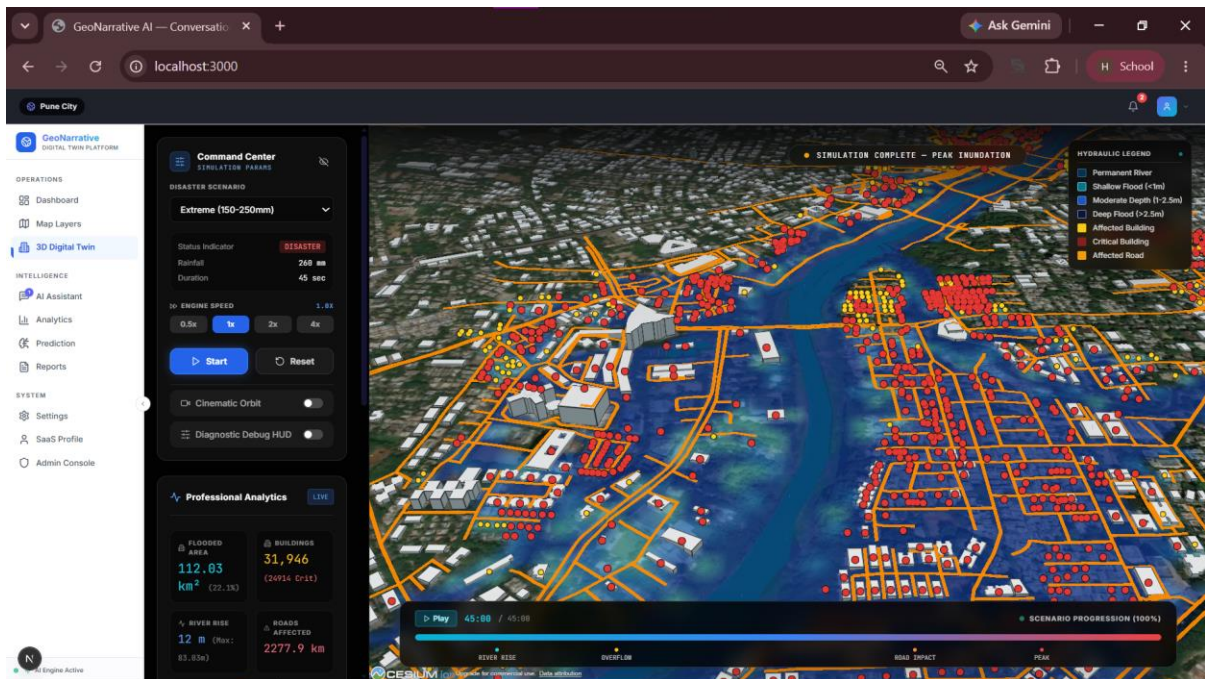


Figure 6.16: 3D Digital Twin View capturing animated temporal flood progression, Red critical building overlays, and flooded road corridors.

6.9.4 Conversational GeoAI Assistant

The natural language decision support interface demonstrates real-time conversational GeoAI capabilities powered by Google Gemini. When tasked with administrative inquiries such as 'Identify total road disruption and critical building impacts under extreme flood scenarios in Pune,' the AI Assistant extracts parameters from active simulation stores and responds with quantitative analytical intelligence. Using autonomous tool routing, the copilot executes map command shifts and spatial filter adjustments instantly upon user request.

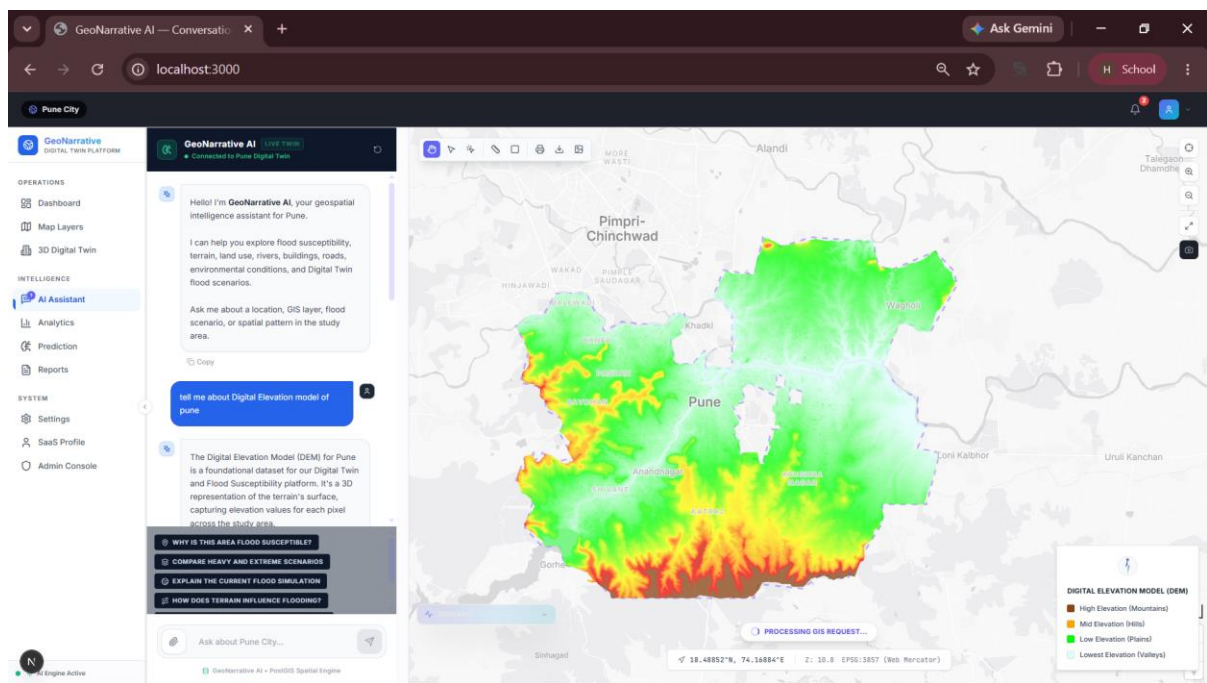


Figure 6.17: Interactive GeoAI Assistant copilot panel

6.9.5 Technical Report Agent Studio

Complementing the AI copilot, the Geospatial Technical Report Agent studio generates publication-grade structural risk reports. Configured directly from live application state telemetry, the Report Agent compiles comparative scenario statistics into formatted dossiers incorporating vulnerability matrices and mitigation guidelines, bridging software computations with formal municipal governance workflows.

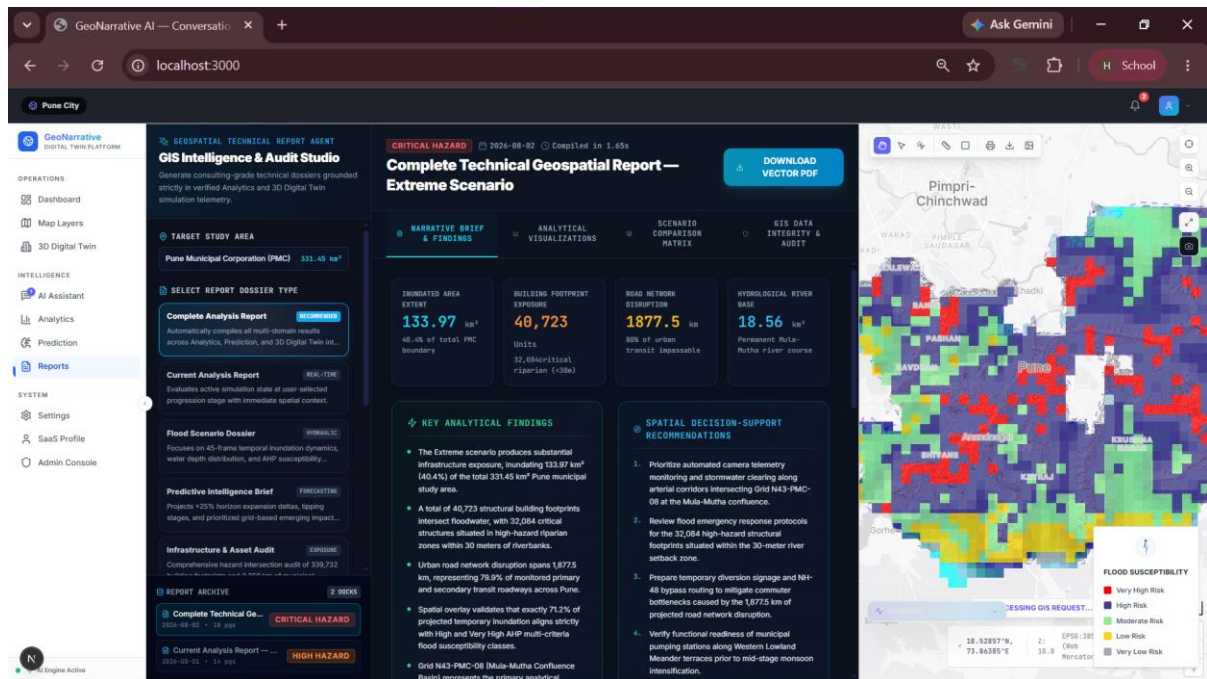


Figure 6.18: automated Technical Report Agent studio generating disaster briefings.

6.9.6 Spatial Analytics Dashboard

The Spatial Analytics Dashboard consolidates the principal geospatial outputs of GeoNarrative into an interactive analytical workspace for interpreting flood susceptibility and urban exposure across Pune. It integrates the classified susceptibility surface with summary indicators for buildings, road networks, water features, and risk-zone distribution, allowing spatial patterns to be examined alongside their corresponding quantitative measures. The dashboard transforms individual GIS layers into a unified decision-support view, enabling rapid comparison of vulnerability classes and infrastructure exposure while maintaining direct geographic context on the map.

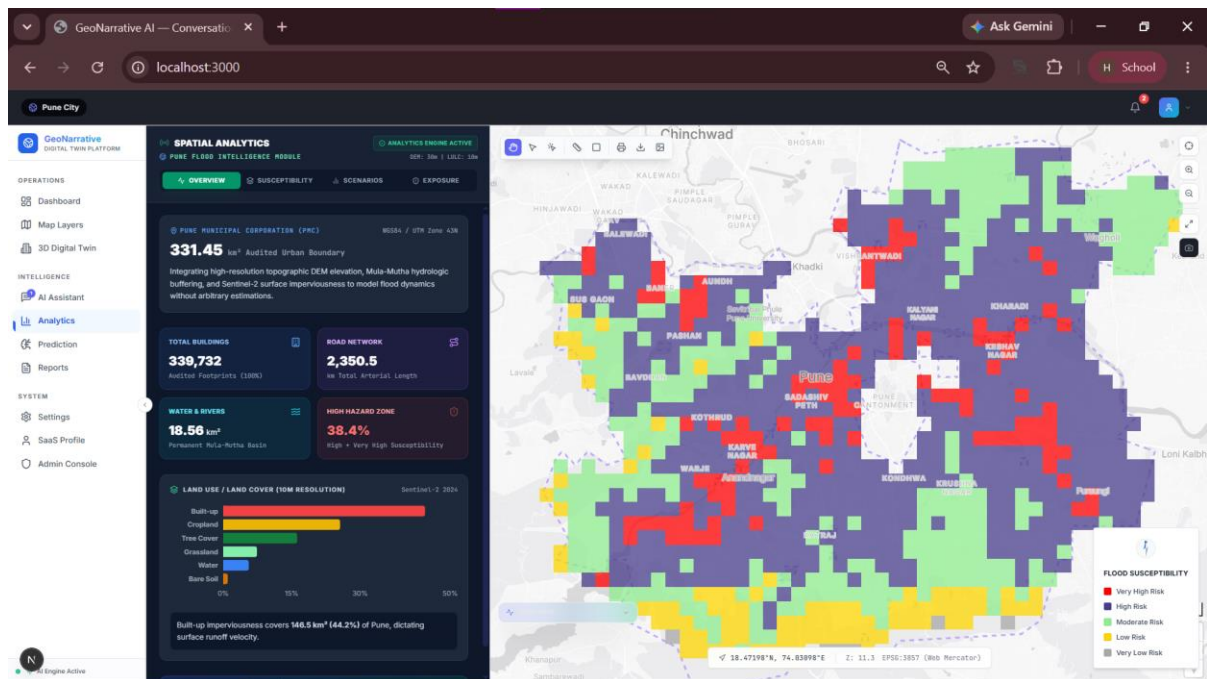


Figure 6.19: Spatial Analytics Dashboard integrating flood susceptibility, infrastructure exposure, and risk-distribution statistics.

6.9.7 Predictive Analytics and Scenario Intelligence Dashboard

The Predictive Analytics Dashboard extends the susceptibility assessment by providing scenario-oriented intelligence for anticipating changes in flood exposure under varying hazard conditions. It combines spatial risk patterns with scenario controls and projected impact indicators to examine how susceptible areas, exposed buildings, and infrastructure may respond as flood severity increases. Rather than representing a calibrated hydrodynamic forecast, the module functions as a scenario-based decision-support environment, translating the underlying geospatial analysis into interpretable risk projections that can support preparedness, prioritisation, and comparative assessment of potential flood impacts across Pune.

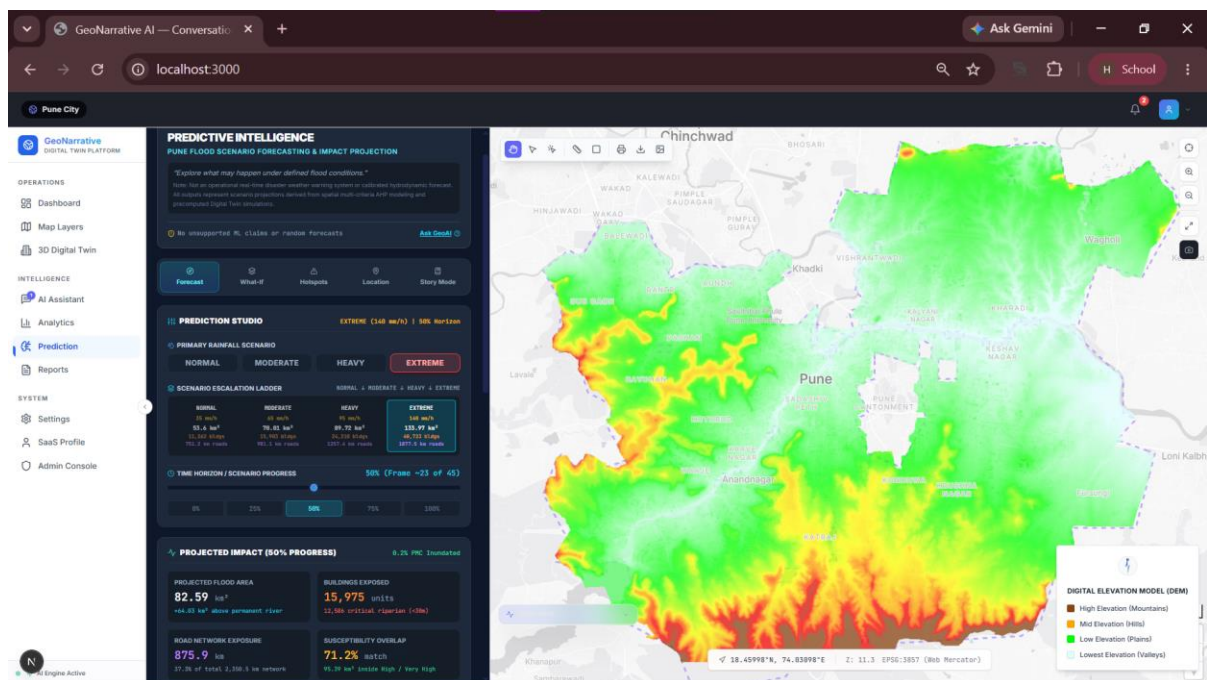


Figure 6.20: Predictive Analytics Dashboard presenting scenario-based flood-risk projections and spatial decision-support indicators

6.10 Scholarly Discussion of Findings

The analytical and system-level results support the central proposition of this thesis: coupling multi-criteria GIS analytical modeling with dynamic 3D Digital Twin visualization and conversational AI extends conventional geospatial decision support. Whereas conventional static susceptibility mapping stops at delineating potential hazard zones, the implemented temporal scenario pipeline quantifies explicit structural vulnerabilities, showing the progression from 11,262 affected buildings under Normal conditions to 40,723 affected structures under Extreme conditions.

The integration of natural-language AI processing is designed to reduce technical proficiency barriers associated with conventional GIS interfaces by enabling users to interrogate spatial models through conversational interactions. In this way, GeoNarrative AI supports broader accessibility of complex hazard intelligence and presents urban disaster-risk information through a dynamic, interactive decision-support environment.

6.11 Chapter Summary

This chapter presented the empirical findings of the GeoNarrative AI framework. Spatial multi-criteria analysis established a five-tier flood susceptibility map across the 506.91 km² PMC territory, showing qualitative spatial coherence with documented historical flood-affected corridors. Temporal scenario simulations quantified computed structural disaster exposure across 180,307 buildings and 7,445.90 km of roadway, demonstrating enforced monotonic scenario expansion up to 133.97 km² of computed inundation extent and 40,723 affected buildings under extreme conditions. Finally, interactive application demonstrations verified the integration of 3D Digital Twin animations, conversational GeoAI assistance, and automated technical reporting. Chapter 7 extends these findings into a critical evaluation of methodological insights, contributions to spatial science, and practical municipal applications.

Chapter 7: Conclusion and Future Directions

7.1 Summary of Research Achievements

This research successfully designed, developed, and evaluated GeoNarrative AI, an advanced spatial decision-support system (SDSS) coupling multi-criteria GIS flood modeling, terrain-constrained scenario simulations, 3D Digital Twin visualization, and domain-grounded conversational GeoAI for the Pune Municipal Corporation (PMC). The research bridged classical barriers separating static spatial analytics from interactive decision support through four primary deliverables:

- A Multi-Criteria Flood Susceptibility Model covering the exact 506.91 km² territorial extent of the Pune Municipal Corporation, formulated via a 5-factor AHP-informed weighted overlay (Elevation, Distance to Waterways, Slope Gradient, LULC Impermeability, and Building Density).
- A Terrain-Constrained Temporal Scenario Propagation Engine simulating spatial flood expansion across four scenario-based severity levels (Normal, Moderate, Heavy, and Extreme) adhering to enforced monotonic spatial growth constraints.
- Quantitative Vector Infrastructure Exposure Intersections evaluating computed flood exposure across 180,307 municipal building footprints and 55,309 road network segments (7,445.90 km cumulative baseline distance).
- An Integrated Enterprise Web Platform operating over CesiumJS 3D rendering, PostGIS geodatabases, FastAPI spatial microservices, Google Gemini-powered conversational GeoAI Assistant tool routing, and an automated Technical Report Agent.

7.2 Verification of Objective Achievements

A systematic verification of research objectives against empirical deliverable outputs demonstrates the implementation of the defined research objectives, as detailed in Table 7.1.

Table 7.1: Research Objective Achievement Verification Matrix

Research Objective	Operational Status	Empirical Evidence & Chapter Trace
Objective 1: Construct multi-criteria FSM for PMC area	Achieved	5-class FSI susceptibility raster and zonal classification across 506.91 km ² municipal extent (Chapter 4.4, Chapter 6.7, Figure 6.10).
Objective 2: Simulate temporal flood scenarios and evaluate infrastructure exposure	Achieved	Scenario metrics table quantifying building impacts (11,262 up to 40,723 units) and road disruption (751.19 up to 1,877.47 km) (Chapter 4.5–4.6, Chapter 6.8, Table 6.2).
Objective 3: Develop interactive 3D Digital Twin with GeoAI copilot and reporting	Achieved	CesiumJS 3D globe with hazard rendering (Red <30m buffers), conversational GeoAI tool agent, and Technical Report Studio (Chapter 5, Chapter 6.9, Figures 6.12–6.20).

7.3 Answers to Research Questions

RQ1: How can multi-criteria GIS spatial conditioning factors be integrated to delineate relative flood susceptibility zones across Pune Municipal Corporation?

Spatial susceptibility across urban monsoon environments is effectively stratified by integrating five primary topographical, hydrological, and anthropogenic conditioning factors through an AHP-informed weighted-overlay multi-criteria evaluation structure. Assigning hierarchical dominance to Copernicus DEM Elevation (35%) and Distance to Mula-Mutha Waterways (25%), supported by Terrain Slope (20%), Sentinel-2 LULC Impermeability

(10%), and Building Density (10%), was used to represent the selected spatial flood-conditioning relationships without requiring dense historical gauge networks. The resulting categorical susceptibility classification demonstrated qualitative spatial coherence with documented 2019 and 2024 flood disaster corridors along central riverine wards.

RQ2: How can scenario-driven temporal flood propagation be modeled and coupled with spatial vector intersections to quantify progressive infrastructure exposure under varying scenario severity levels?

Temporal flood dynamics can be modeled without excessive fluid dynamic computational penalties by deploying a terrain-constrained propagation algorithm driven outward from permanent water geometries (171 hydrographic features). Enforcing strict mathematical monotonicity ($A(t_1) \leq A(t_2) \leq \dots \leq A(t_n)$) preserves hydrological visual logic across simulation frames. Executing vector-raster spatial intersections (ST_Intersects in UTM Zone 43N) quantified a monotonic increase in computed municipal infrastructure exposure: inundated surface areas expand from 53.60 km² in Normal scenarios up to 133.97 km² under the Extreme scenario, concurrently driving critical riparian structural exposure within 30-meter river buffers from 8,808 up to 32,084 building structures, and expanding roadway submergence from 751.19 km to 1,877.47 km.

RQ3: How can an integrated 3D Urban Digital Twin and AI-assisted geospatial decision-support environment support the interpretation, communication, and exploration of spatial flood intelligence?

Transitioning spatial analytics from static two-dimensional cartography into an interactive CesiumJS 3D Digital Twin is designed to support human risk comprehension by visualizing flood progression against volumetric urban morphology and topographic relief; as no formal usability study (e.g., task-completion timing or SUS survey) was conducted, this is presented as a design rationale rather than an empirically measured comprehension gain. Coupling the digital twin with a generative conversational GeoAI Assistant (powered by Google Gemini) is intended to lower traditional software technical barriers; administrative authorities can query hazard geodatabases via natural-language dialogues and automated tool executions. Furthermore, the automated Technical Report Agent synthesizes simulation telemetry into

structured engineering briefings, offering a workflow that connects quantitative geospatial modeling with municipal disaster governance reporting.

7.4 Synthesis Alignment Matrix

Table 7.2: Synthesis Alignment Matrix of Research Questions, Objectives, Methods, and Chapters

Research Question	Corresponding Objective	Primary Methodological Contribution	Chapter Documentation Trace
RQ1: Spatial susceptibility formulation & MCDA	Objective 1: Multi-criteria AHP-informed model	Parsimonious 5-factor AHP-informed weighted overlay over Copernicus DEM and Sentinel-2 LULC	Chapter 4 (Sec 4.4), Chapter 6 (Sec 6.7)
RQ2: Temporal scenario propagation & exposure	Objective 2: Scenario infrastructure evaluation	Terrain-constrained monotonic raster expansion and PostGIS vector spatial intersections	Chapter 4 (Sec 4.5–4.6), Chapter 6 (Sec 6.8)
RQ3: 3D Digital Twin & GeoAI decision support	Objective 3: Interactive SDSS with GeoAI	Asynchronous FastAPI/PostGIS backend, CesiumJS 3D frontend, Gemini GeoAI copilot, & Report Agent	Chapter 4 (Sec 4.8–4.9), Chapter 5, Chapter 6 (Sec 6.9)

7.5 Methodological Limitations and Caveats

To ensure scientific academic transparency, several methodological parameters and boundaries are acknowledged as formal research limitations:

- **DEM Horizontal Resolution:** The 30-meter horizontal grid structure of the Copernicus GLO-30 DEM smooths out micro-scale urban topological features—such as stormwater drainage culverts, underpass tunnels, road embankments, and retaining walls—that govern localized pluvial flow paths in dense Peth wards.

- **Inundation Depth Artifacts:** As explicitly documented in Chapter 4 and Chapter 6, maximum simulated water depth values represent scenario-derived relative DEM elevation differences over flooded cells rather than field-verified shallow water hydrograph measurements. Local maximum outliers (e.g., 89.38m in extreme datasets) emerge from vertical DEM sink anomalies and steep canyon sidewalls within discrete raster grid columns.
- **Adoption of Literature Weights:** Criterion weights within the AHP-informed weighted overlay were adapted from published urban flood-susceptibility literature rather than derived via an independent expert pairwise-comparison matrix specific to municipal engineering officers in Pune; consequently, no original AHP Consistency Ratio (CR) was computed, and no CR value was carried over from the source literature.
- **Simulation Architecture vs. Hydrodynamic Equations:** The temporal flood propagation engine utilizes terrain-constrained geographic neighborhood expansion to guarantee real-time interactive 60-FPS UI rendering, rather than computationally resolving 2D Saint-Venant shallow water fluid dynamics (e.g., fully calibrated HEC-RAS 2D).
- **Building Structural Proxies:** OpenStreetMap architectural polygon vectors supply footprint bounds and estimated structural heights but lack quantitative attributes regarding basement construction, first-floor elevation height, or structural material strength, limiting damage valuation estimations.

7.6 Recommendations for Future Directions

To build upon the technological foundations established by GeoNarrative AI, several avenues for future research and empirical enhancement are recommended:

- **Integration of Airborne LiDAR Topography:** Replace the satellite-derived 30m DEM with sub-meter resolution airborne LiDAR elevation point clouds and incorporate municipal underground stormwater pipeline schematics to enable ultra-high-fidelity micro-urban pluvial modeling.
- **Quantitative ROC-AUC Hydrodynamic Validation:** Obtain empirical radar satellite imagery (e.g., Sentinel-1 SAR flood extent footprints) captured during active historical inundation events to conduct quantitative area-under-the-curve (ROC-AUC) spatial accuracy validations against susceptibility layers.

- **Real-Time Telemetry & IoT Weather Ingestion:** Connect the backend FastAPI event broker directly to automated India Meteorological Department (IMD) rainfall gauges and municipal river-level telemetry stations, enabling automated real-time scenario initialization and live twin updates during active monsoon storms.
- **Hybrid Physics-Informed Neural Networks (PINNs):** Integrate machine learning physical surrogate models (such as physics-informed deep operator networks) within the simulation backend to approximate real-time 2D Navier-Stokes fluid dynamics without compromising frontend rendering speed.
- **Socio-Economic & Demographic Vulnerability Indexing:** Enrich structural vector intersections with Ward-level demographic census datasets, property economic valuations, and critical public health asset inventories to formulate a comprehensive socio-economic disaster risk index.
- **Extended GeoAI Autonomous Action Spaces:** Expand the Google Gemini tool agent interface to automate real-time emergency broadcast dissemination, compute safe multi-modal evacuation routing around inundated road networks, and autonomously dispatch technical vulnerability reports to designated municipal responder channels.

7.7 Concluding Remarks

The GeoNarrative AI dissertation demonstrates that uniting multi-criteria spatial analytical modeling, temporal 3D Digital Twin visualization, and domain-grounded conversational artificial intelligence contributes a practical, technically feasible approach to urban disaster decision support. By executing precise quantitative evaluations across 180,307 structures and 7,445.90 kilometers of transportation lifeline across the Pune Municipal Corporation territory, By executing quantitative spatial evaluations across 180,307 structures and 7,445.90 kilometres of transportation networks across the Pune Municipal Corporation territory, the framework translates spatial modelling assumptions into reproducible scenario-based risk intelligence. In addition, integrating conversational GeoAI directly into the spatial exploration loop is designed to reduce technical accessibility barriers, transforming static GIS databases into dynamic, intuitive, and actionable instruments for smart city disaster resilience and operational urban governance.

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